

HR Analytics Dashboard

Complete technical documentation of a 7-worksheet Tableau attrition dashboard, extracted from the workbook XML

DASHBOARD	HR Dashboard (the name carries a trailing space)
WORKBOOK	HR Analytics Dashboard.twb (packaged workbook)
AUTHOR	Atharva Devne
PROJECT TYPE	Self-directed, for the author's own skill development
DATASET	Kaggle, public dataset. HR_data.csv, 39 columns, 15 consumed
TABLEAU	Desktop 2025.2.0 (20252.25.0514.2217), macOS, doc format 18.1
CANVAS	1580 x 900 px, sizing-mode='fixed', floating over a background image
SCALE	7 worksheets, 1 dashboard, 6 calculated fields, 1 parameter, 6 actions
STATUS	Published to Tableau Public, revision 1.1

SCREENSHOT SLOT

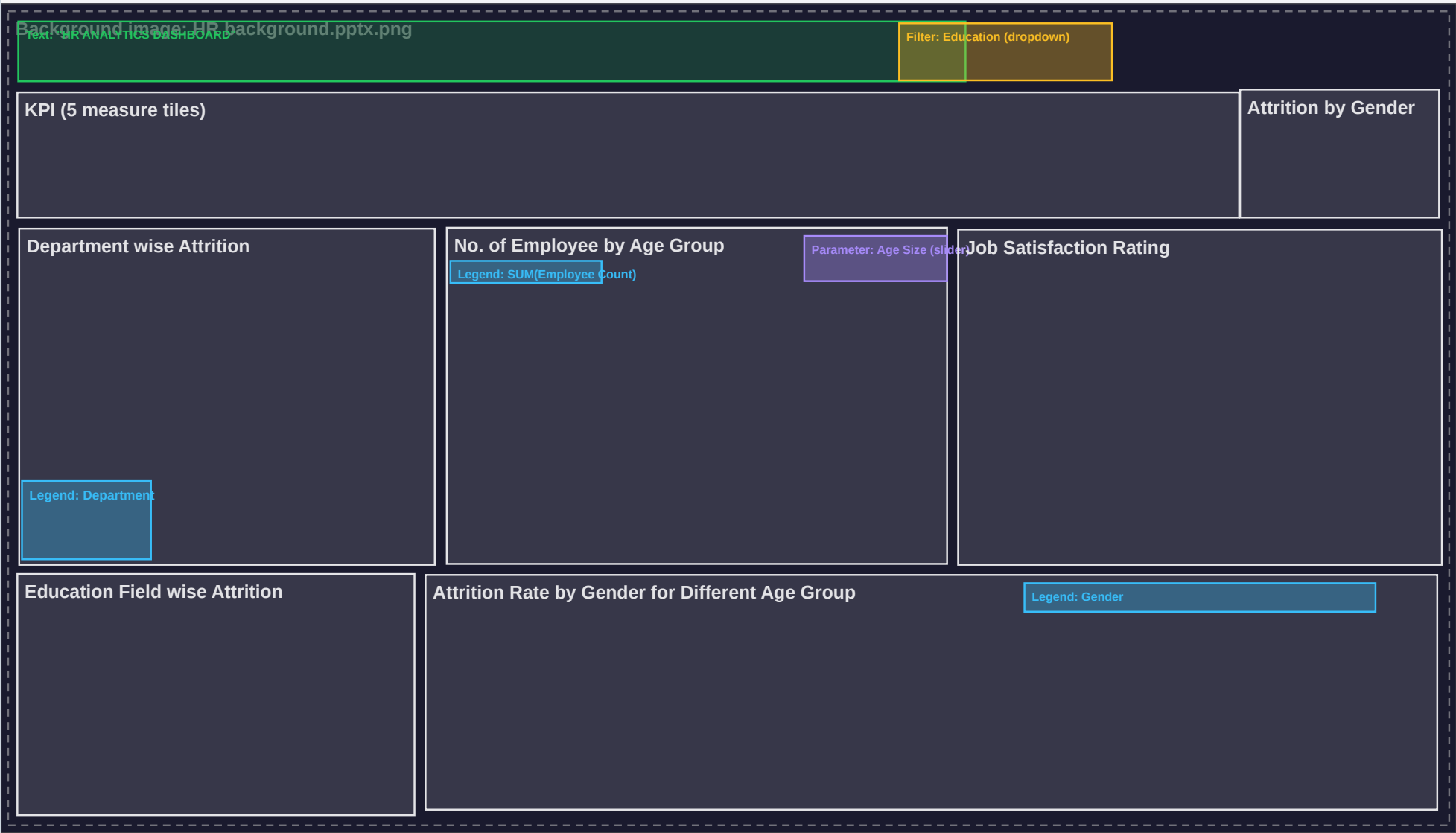
Paste your own dashboard screenshot here. No real screenshot could be captured: no Tableau runtime was available and the network policy denied public.tableau.com.

TABLEAU PUBLIC LINK

Recoverable from the XML, to be verified by you: workbook id HRAnalyticsDashboard_17561606616710, dashboard path /workbooks/HRAnalyticsDashboard_17561606616710/HRDashboard, current revision 1.1.

Dashboard 2 - HR Dashboard (1580 x 900 px, fixed sizing)

Coordinate-accurate wireframe reconstructed from <zones> in the .twb XML. Not a screenshot.



Coordinate-accurate wireframe reconstructed from the <zones> geometry in the .twb XML. Not a screenshot.

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PART A

Full technical documentation

Sections 1 to 19: overview, data sources, data model, preparation, field inventory, calculated fields, parameters, filters, KPIs, worksheet analysis, visualization breakdown, layout, actions, tooltips, business logic, enabled insights, technical summary, interview explanation and resume bullets.

1. DASHBOARD OVERVIEW

Item	Value	Evidence
Dashboard name	HR Dashboard	✓ DIRECTLY VERIFIED. Note the trailing space. The XML is <code><dashboard name='HR Dashboard '></code> and every action targets <code>value="HR Dashboard "</code> with that space. It is part of the actual object name.
Workbook name	HR Analytics Dashboard.twb (inside a packaged archive)	✓ DIRECTLY VERIFIED
Tableau version	Document format <code>version='18.1'</code> , <code>original-version='18.1'</code> ; <code>source-build='2025.2.0 (20252.25.0514.2217)'</code> ; build comment <code>20252.25.0723.1135</code>	✓ DIRECTLY VERIFIED
Source platform	<code>source-platform='mac'</code>	✓ DIRECTLY VERIFIED
Publish target	<code>repository-location derived-from='https://public.tableau.com/workbooks/HRAnalyticsDashboard_17561606616710?rev=1.0'</code> , <code>id='HRAnalyticsDashboard_17561606616710'</code> , <code>revision='1.1'</code> . The dashboard itself: <code>id='HRDashboard'</code> , <code>path='/workbooks/HRAnalyticsDashboard_17561606616710'</code>	✓ DIRECTLY VERIFIED
xml:base	<code>https://public.tableau.com</code>	✓ DIRECTLY VERIFIED
Author / owner	Atharva Devne. Confirmed by the author. The workbook itself stores no author field.	✓ CONFIRMED BY AUTHOR (not from the file)
Author evidence in the file	Source file paths are <code>/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main</code> , so the macOS account was <code>atharvadevne</code> . The <code>-main</code> suffix is the naming convention produced by downloading a GitHub repository ZIP of a branch called <code>main</code> .	✓ VERIFIED path; the <code>-main</code> reading is △ INFERRED

Item	Value	Evidence
Dashboard dimensions	1580 x 900 px, <code>sizing-mode='fixed'</code>	✓ DIRECTLY VERIFIED
Dashboard UUID	<code>{30867C40-EA91-4DA5-AF47-E92ACECA09BD}</code>	✓ DIRECTLY VERIFIED
Dashboard -level style	<code>parameter-ctrl</code> text <code>#ffffff</code> ; <code>parameter-ctrl-title</code> text <code>#f1ce63</code> in <code>Tableau Semibold</code>	✓ DIRECTLY VERIFIED

Dashboards contained in the workbook

#	Dashboard name	Analyzed here
1	<code>HR Dashboard</code>	Yes. It is the only dashboard in the workbook.

Stories: none. ✓ DIRECTLY VERIFIED

Purpose, business problem, audience, story

Unlike Dashboard 1, this workbook does contain one piece of authored text: the dashboard title text object reads `HR ANALYTICS DASHBOARD` (zone id 7). ✓ DIRECTLY VERIFIED. Beyond that there is no description, no annotation and no caption. ✗ NOT AVAILABLE as stated fact.

What can be read off the structure (△ INFERRED):

- **Purpose (inferred).** Attrition analysis. `Attrition Count` is the measure behind five of the seven worksheets, and four of the seven worksheet names contain the word "Attrition".
- **Business question (inferred).** "Who is leaving the organisation, from which departments, education fields, age bands and genders, and how does that sit against headcount and job satisfaction?"
- **Project type (confirmed by the author). Self-directed. Built for the author's own skill development,** not for a course, a client or an employer.
- **Dataset origin (confirmed by the author). Kaggle.** A public dataset.
- **Intended audience (inferred).** The design is framed for HR business partners, a people analytics function or HR leadership: the KPI strip leads with headcount and attrition, which is the standard opening for an HR leadership view. In practice the audience is anyone reviewing the author's portfolio.
- **Overall story (inferred).** A title bar with one global Education filter, then a five-tile KPI strip with a companion gender breakdown, then three mid-canvas views splitting attrition by department, headcount by age band and satisfaction by job role, then two lower views splitting attrition by education field and by gender within age band.

Every chart is click-to-filter, so the dashboard functions as an exploratory cross-filtering surface rather than a fixed report.

2. DATA SOURCE INVENTORY

This workbook contains **five** `<datasource>` elements. One is the parameter container, four are real connections, and **only one of the four is actually used**. Each is documented separately below.

Data Source	Type	Connection	Tables/Files	Important Fields	Live/Extract	Notes
Parameters	Parameter container (Tableau built-in, not a real connection)	<code>hasconnection='false'</code>	none	[Age Parameter] (caption Bin Size)	n/a	Holds the single parameter in the workbook
HR data (HR Data) (federated.0f0h03p1it3gvo176rhlnly1tx4v)	Federated over a flat file	Two named connections declared: (a) <code>excel-direct</code> named HR Data pointing at <code>/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main/HR data.csv</code> ; (b) <code>textscan</code> named HR_data pointing at directory <code>/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main, filename HR_data.csv</code> . The active relation uses the textscan connection.	1 table: [HR_data#csv]	Attrition, Employee Count, Department, Education Field, Gender, Job Role, Job Satisfaction, Age, CF_age band	Extract , <code>enabled='true'</code> , <code>count='-1'</code> , <code>units='records'</code>	This is the only data source any worksheet uses. All 7 worksheets and the dashboard bind to it.
HR Data.xlsx - HR data (federated.049m4yg1psanb61b4tu3001365fp)	Federated over a flat file	<code>textscan</code> , directory <code>/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main</code> , filename <code>HR Data.xlsx - HR data.csv</code>	1 table: [HR Data.xlsx - HR data#csv]	Same 39-column schema	Extract , <code>count='-1'</code>	Unused. No worksheet references it.
HR data (federated.1g91kqr1twrfkg14hlu c71es07jw)	Federated over a flat file	<code>textscan</code> , same directory, filename <code>HR data.csv</code>	1 table: [HR data#csv]	Same 39-column schema	Extract , <code>count='-1'</code>	Unused.
HR data (2) (federated.1u260d51era62r18fltmd02kg202)	Federated over a flat file	<code>textscan</code> , same directory, filename <code>HR data.xlsx</code>	1 table: [HR data.xlsx]	Same 39-column schema	Extract , <code>count='-1'</code>	Unused.

✓ DIRECTLY VERIFIED for every cell.

2.1 Notes on the four connections

All four point at the same folder and the same 39-column schema. The four filenames differ (`HR_data.csv` , `HR data.csv` , `HR data.xlsx` , `HR Data.xlsx` - `HR data.csv`), which is the signature of **repeated import attempts of the same dataset in different file formats and naming conventions** before settling on one. That is an observation, not a criticism. $\triangle$ INFERRED from the naming pattern; the XML does not record intent.

Why this matters for the packaged archive: all four data sources have extracts enabled, which is exactly why this `.twbx` carries **four** `.hyper` files of 196,608 bytes each in `Data/tableau-temp/` , three of which are dead weight. ✓ VERIFIED (the archive listing).

The `excel-direct` named connection is worth flagging. The active data source declares a connection of `class='excel-direct'` whose `filename` is `/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main/HR_data.csv` , that is, the Excel connector pointed at a `.csv` path. Its attributes are `cleaning='no'` , `compat='no'` , `interpretationMode='0'` , `validate='no'` , `dataRefreshTime=''` . It is declared but the `<relation>` element binds to the `textscan` connection, not this one. ✓ VERIFIED.

2.2 Active data source, detailed attributes

Attribute	Value
Caption	HR data (HR Data)
Internal name	<code>federated.0f0h03plit3gvo176rhln1y1tx4v</code>
<code>inline</code>	<code>true</code> (embedded)
Active connection	<code>textscan.0ry86yc0wlh17j1d0atub17l183k</code>
File	<code>HR_data.csv</code>
Directory	<code>/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main</code>
Relation	<code><relation name='HR_data.csv' table='[HR_data#csv]' type='table' /></code>
Number of tables	1
Source columns	39
Database / schema	None (flat file)
Extract	<code>enabled='true'</code> , <code>count='-1'</code> , <code>units='records'</code> , <code>user-specific='false'</code>
Extract relation	<code><relation name='Extract' table='[Extract].[Extract]' type='table' /></code> under <code><properties context='extract'></code>
Data source filters	One. A <code><shared-view></code> filter on <code>[none:Education:nk]</code> , see section 8.
Extract filters	None.
Custom SQL	None. <code>type='table'</code> , no SQL text.
Aliases	None. No <code><aliases></code> element exists anywhere in this workbook.

Attribute	Value
Renamed fields	Yes , 10 fields carry captions, 6 of which are the <code>Name1</code> pattern. See section 4.
Hidden fields	Yes , 4 auto-generated hidden action groups. See section 5.
Geographic roles	None . No map is used.

2.3 The 39 source columns (identical across all four data sources)

#	Column	Datatype	#	Column	Datatype
0	Attrition	string	20	Environment Satisfaction	integer
1	Business Travel	string	21	Hourly Rate	integer
2	CF_age band	string	22	Job Involvement	integer
3	CF_attrition label	string	23	Job Level	integer
4	Department	string	24	Job Satisfaction	integer
5	Education Field	string	25	Monthly Income	integer
6	emp no	string	26	Monthly Rate	integer
7	Employee Number	integer	27	Num Companies Worked	integer
8	Gender	string	28	Percent Salary Hike	integer
9	Job Role	string	29	Performance Rating	integer
10	Marital Status	string	30	Relationship Satisfaction	integer
11	Over Time	string	31	Standard Hours	integer
12	Over18	string	32	Stock Option Level	integer
13	Training Times Last Year	integer	33	Total Working Years	integer
14	Age	integer	34	Work Life Balance	integer
15	CF_current Employee	integer	35	Years At Company	integer
16	Daily Rate	integer	36	Years In Current Role	integer
17	Distance From Home	integer	37	Years Since Last Promotion	integer
18	Education	string	38	Years With Curr Manager	integer
19	Employee Count	integer			

The three CF_ columns matter. `CF_age band`, `CF_attrition label` and `CF_current Employee` carry the `CF_` prefix that Tableau uses when a **calculated field is exported to a flat file**. They arrive as ordinary source columns here, which means **that derivation was done upstream, outside this workbook**, and this workbook consumes the result. `CF_age band` in particular is used directly on a shelf without any calculation in this workbook. △ INFERRED from the naming convention; ✓ VERIFIED that they are plain source columns here with no calculation attached.

3. DATA MODEL / RELATIONSHIPS / JOINS

Each of the four real data sources is a single flat table. There is no data model. ✓ DIRECTLY VERIFIED

Check	Result
Joins	None. Each <code><connection></code> holds exactly one <code><relation type='table'></code> .
Relationships (noodles)	None. The <code><object-graph></code> contains a single object per data source.
Unions	None.
Blending	None , and this is worth stating precisely. Four data sources exist, which is the usual precondition for blending, but no worksheet references more than one of them. There is no <code><datasource-dependencies></code> in any worksheet pointing at two data sources, and no linking field is defined. The other three data sources are simply inert.
Nested joins	None.
Cardinality / referential integrity	Not applicable.
Custom SQL	None.

Plain English

The workbook reads one CSV, `HR_data.csv`, as a single flat table of 39 columns at one row per employee. Tableau wraps it in a federated container and extracts it whole to Hyper. Three other data sources point at variants of the same file and were left in the workbook without ever being used. Every worksheet and the dashboard bind to the single active source, plus the `Parameters` container for the one parameter. There is nothing to join, relate, union or blend.

Text representation

```
HR_data.csv    (39 columns, one row per employee)
  |
  |  textscan connection, full extract (count = -1)
  v
federated.0f0h03plit3gvo176rhln1y1tx4v  "HR data (HR Data)"  [ACTIVE]
  |
  |--> shared-view data source filter on [Education]
  |--> 7 worksheets
  |--> HR Dashboard

Parameters  (no connection)
  |
  |--> [Age Parameter] "Bin Size"
        |
        |--> size-parameter of [Age (bin)]
        |--> slider control on HR Dashboard

HR Data.xlsx - HR_data.csv --> federated.049m4yg1psanb61b4tu3001365fp  [UNUSED, extract present]
HR_data.csv   --> federated.1g91kqr1twrfkg14hluc71es07jw  [UNUSED, extract present]
HR_data.xlsx  --> federated.1u260d51era62r18fltmd02kg202  [UNUSED, extract present]
```

4. DATA PREPARATION / CLEANING

4.1 Field renaming (captions)

Ten fields in the active data source carry captions. Six of them follow the `Name1` pattern.

Raw column	Caption applied	Comment
Department	Department1	
Education	Education1	
Education Field	Education Field1	
Gender	Gender1	
Job Role	Job Role1	
Job Satisfaction	Job Satisfaction1	Also role-converted, see 4.2
emp no	Emp No	Genuine readability rename
CF_age band	CF age band	Underscore removed
CF_attrition label	CF attrition label	Underscore removed

Raw column	Caption applied	Comment
CF_current Employee	CF current Employee	Underscore removed

Honest reading of the `Name1` pattern (Δ INFERRED). A trailing `1` is what Tableau appends automatically when a field name would collide with an existing one, typically after replacing or re-adding a data source. These are almost certainly **not intentional business renames**. They are collision artefacts that were left in place. This is worth knowing because a hiring manager reading `Department1` in a legend will ask about it. Note that the **tooltips override the display**: the custom tooltips spell out `Department:`, `Gender:`, `Job Role:`, `Education Field:` and `Job Satisfaction:` as hand-typed labels, so the `1` does not leak into the tooltip text. ✓ VERIFIED from the tooltip XML.

4.2 Role conversion

Field	Role	Type	Note
[Job Satisfaction]	dimension	ordinal	Converted from measure. <code>aggregation='Sum'</code> retained as the default. Used as a discrete column header on the highlight table.
[Employee Number]	dimension	ordinal	Converted from measure, so it is not summed.
[Age]	measure	quantitative	Left as a measure, used with <code>AVG</code> on the KPI strip and as the input to the bin
[CF_current Employee]	measure	quantitative	Left as a measure. Not used on any sheet.

4.3 Parameterised binning

Bin field	Source field	Bin size
[Age (bin)]	[Age]	<code>size-parameter='[Parameters].[Age Parameter]'</code>

Exact definition: `<calculation class='bin' decimals='0' formula='[Age]' peg='0' size-parameter='[Parameters].[Age Parameter]'` />

What was done: `Age` was binned, but instead of a hard-coded width the bin size is bound to a parameter, so the end user can change the bin width at view time from a slider on the dashboard. **Why it was likely necessary (Δ inferred):** an age histogram reads very differently at 2-year buckets versus 10-year buckets. Letting the user choose avoids committing to one granularity. **This is the single most advanced data-preparation technique in either workbook.** ✓ VERIFIED.

Contrast with Dashboard 1, where all seven bins are fixed-size.

4.4 Cleaning calculated fields

See section 6 for full detail. In summary:

Calculated field	Formula	Cleaning function
Attrition Count	IF [Attrition] = 'Yes' THEN 1 ELSE 0 END	Converts a Yes/No string column into a numeric 0/1 flag so it can be aggregated. This is a genuine type-and-encoding transformation.

4.5 Transformations that are NOT present

Transformation	Present?
Null handling (ISNULL , IFNULL , ZN)	No
Duplicate handling	No
Explicit type casts (INT() , STR() , FLOAT())	No. The IF/THEN in Attrition Count changes the encoding but is not a cast function.
Date conversions / date parts	No. The dataset has no date field at all . Years At Company , Years In Current Role , Years Since Last Promotion , Years With Curr Manager and Total Working Years are integer tenure counts, not dates.
String manipulation (LEFT , SPLIT , TRIM , REPLACE , CONTAINS)	No
Split fields	No
Pivot	No
Union	No
Value aliases	No. Unlike Dashboard 1, this workbook uses no aliasing at all.
Groups (data groups)	No. The four <group> elements present are auto-generated hidden action groups (user:auto-column='sheet_link'), not user-created groups.
Sets	No
Hierarchies / drill paths	No
Geographic roles	No
Tableau Prep flow	No
Extract filter	No
Upstream derivation	Partly, but outside this workbook. The CF_age band , CF_attrition label and CF_current Employee columns arrive already derived.

5. COMPLETE FIELD INVENTORY

5.1 Fields declared in the active data source (18 <column> elements)

Field Name	Original Name	Data Type	Role	Aggregation	Table	Description / Usage
Age (bin)	derived from [Age]	integer	Dimension (ordinal)	None	HR_data.csv	Columns shelf of No. of Employee by Age Group . Bin width driven by the Bin Size parameter.

Field Name	Original Name	Data Type	Role	Aggregation	Table	Description / Usage
Age	Age	integer	Measure	(default)	HR_data.csv	AVG(Age) on the KPI strip; input to Age (bin)
CF age band	CF_age band	string	Dimension (nominal)	None	HR_data.csv	Columns shelf of Attrition Rate by Gender for Different Age Group; manually sorted
CF attrition label	CF_attrition label	string	Dimension (nominal)	None	HR_data.csv	Declared but not used on any worksheet
CF current Employee	CF_current Employee	integer	Measure	None	HR_data.csv	Declared but not used on any worksheet
Attrition Count	[Calculation_231935388807032832]	integer	Measure	Sum in use	HR_data.csv	The core measure. Drives 5 of the 7 worksheets
Attrition Rate	[Calculation_231935388807876609]	real	Measure	User-defined aggregate	HR_data.csv	KPI strip only
Active Employees	[Calculation_231935388808294402]	integer	Measure	User-defined aggregate	HR_data.csv	KPI strip only
Department1	Department	string	Dimension (nominal)	None	HR_data.csv	Colour of Department wise Attrition; source field of filter Action1
Education Field1	Education Field	string	Dimension (nominal)	None	HR_data.csv	Rows shelf of Education Field wise Attrition; source field of filter Action5
Education1	Education	string	Dimension (nominal)	None	HR_data.csv	The one data source level filter, exposed as a dashboard dropdown
Employee Number	Employee Number	integer	Dimension (ordinal)	None	HR_data.csv	Declared, not used on any worksheet
Gender1	Gender	string	Dimension (nominal)	None	HR_data.csv	Rows/Colour on two worksheets
Job Role1	Job Role	string	Dimension (nominal)	None	HR_data.csv	Rows shelf of Job Satisfaction Rating
Job Satisfaction1	Job Satisfaction	integer	Dimension (ordinal)	Sum	HR_data.csv	Columns shelf of Job Satisfaction Rating
Number of Records	auto-generated	integer	Measure	Sum	HR_data.csv	Formula 1. user:auto-column='numrec'. Not used on any worksheet.
Emp No	emp no	string	Dimension (nominal)	None	HR_data.csv	Declared, not used on any worksheet
HR_data.csv object id	internal	table	Measure	None	HR_data.csv	Tableau internal table object

Plus Employee Count (source column 19), which is **not** given its own <column> override but is used heavily as SUM([Employee Count]).

5.2 Ad-hoc worksheet-level calculated field

Field Name	Internal Name	Formula	Scope	Purpose
min(1)	[Calculation_1302666200262176769]	min(1)	user:unnamed='Attrition Rate by Gender for Different Age Group', that is, created inside that worksheet and never promoted	Axis placeholder for the dual-axis donut construction

5.3 Parameter

Field Name	Caption	Data Type	Domain	Current Value	Range
[Age Parameter]	Bin Size	real	param-domain-type='range'	3.0	min 2.0, max 10.0, granularity 1.0

5.4 Hidden auto-generated action groups (4)

These are created automatically by Tableau when a dashboard filter action is defined. They are marked `hidden='true'` and `user:auto-column='sheet_link'`.

Group name	Caption	Underlying level	Created by
[Action (Age (bin))]	Action (Age (bin))	[Age (bin)]	filter action [Action2]
[Action (Department)]	Action (Department)	[Department]	filter action [Action1]
[Action (Department1)]	Action (Department1)	[Department]	filter action [Action1]
[Action (Education Field)]	Action (Education Field)	[Education Field]	filter action [Action5]

Each has the structure `<groupfilter function='crossjoin'><groupfilter function='level-members' level='[...]' /></groupfilter>`.

5.5 Field categorisation

Category	Members
Dimensions used on the dashboard	Department1, Education Field1, Education1, Gender1, Job Role1, Job Satisfaction1, CF age band, Age (bin)
Measures used on the dashboard	Employee Count (SUM), Attrition Count (SUM), Attrition Rate, Active Employees, Age (AVG), min(1)
Dates	None. The dataset contains no date or datetime field.
Geographic fields	None.
Calculated fields	Attrition Count, Attrition Rate, Active Employees, Number of Records, Age (bin), min(1)
Parameters	[Age Parameter] (Bin Size)
Sets	None.
Groups	4 hidden auto action groups only. No user-created data groups.

Category	Members
Bins	1 (Age (bin) , parameter-driven)

5.6 Usage classification

Classification	Fields
Used on the dashboard	Attrition Count, Attrition Rate, Active Employees, Employee Count, Age, Age (bin), CF age band, Department1, Education Field1, Education1, Gender1, Job Role1, Job Satisfaction1, min(1), [Age Parameter]
Used inside other calculations	Attrition (inside Attrition Count), Attrition Count (inside both Attrition Rate and Active Employees), Employee Count (inside both), Age (inside Age (bin)), [Age Parameter] (inside Age (bin))
Hidden	The 4 auto action groups
Declared but unused	CF attrition label, CF current Employee, Employee Number, Emp No, Number of Records
Source columns never surfaced	24 of the 39 source columns are never used on any shelf, filter or calculation: Business Travel, Marital Status, Over Time, Over18, Training Times Last Year, Daily Rate, Distance From Home, Environment Satisfaction, Hourly Rate, Job Involvement, Job Level, Monthly Income, Monthly Rate, Num Companies Worked, Percent Salary Hike, Performance Rating, Relationship Satisfaction, Standard Hours, Stock Option Level, Total Working Years, Work Life Balance, Years At Company, Years In Current Role, Years Since Last Promotion, Years With Curr Manager. ✓ VERIFIED by absence from every worksheet's <datasource-dependencies>.

6. CALCULATED FIELDS

Calculated Field: Attrition Count

- **Exact Tableau formula:** IF [Attrition] = 'Yes' THEN 1 ELSE 0 END
- **Internal name:** [Calculation_231935388807032832]
- **Data type:** integer
- **Role / type:** measure, quantitative
- **Referenced fields:** [Attrition] (source column 0, string)
- **Used in worksheet(s):** KPI, Attrition by Gender, Department wise Attrition, Education Field wise Attrition, Attrition Rate by Gender for Different Age Group (5 of 7)
- **Used in KPI(s):** the Attrition Count KPI tile, and indirectly the Attrition Rate and Active Employees tiles

- **Purpose:** turn the textual **Yes / No** attrition flag into a numeric indicator so it can be summed.
- **Plain English explanation:** for every employee row, return 1 if that employee left, otherwise 0. Summing this field over any group therefore gives the number of leavers in that group.
- **Business logic:** attrition is defined strictly as **Attrition = 'Yes'**. There is no tenure qualification, no voluntary versus involuntary split, and no exclusion of any status. Whatever the source system meant by **Yes** is what this dashboard counts as attrition.
- **Dependencies on other calculated fields:** none. It is the **root** of the calculation chain.
- **Calculation type:** conditional, **IF/THEN/ELSE**. This is the only conditional logic in either workbook.

Calculated Field: **Attrition Rate**

- **Exact Tableau formula:** **SUM([Calculation_231935388807032832])/SUM([Employee Count])** (written in the Tableau UI as **SUM([Attrition Count])/SUM([Employee Count])**)
- **Internal name:** **[Calculation_231935388807876609]**
- **Data type:** real
- **Role / type:** measure, quantitative
- **Referenced fields:** **[Attrition Count]** (calculated), **[Employee Count]** (source column 19)
- **Used in worksheet(s):** **KPI** only
- **Used in KPI(s):** the **Attrition Rate** tile
- **Purpose:** express attrition as a proportion of headcount rather than an absolute count.
- **Plain English explanation:** add up the leaver flags, add up the employee count, and divide. Because both halves are aggregates, the result is an **aggregate calculation**: it recomputes at whatever level of detail the view is at, rather than averaging row-level ratios. On the KPI tile the view has no dimensions, so it evaluates across the whole filtered population.
- **Business logic:** numerator is leavers, denominator is total employees. **Employee Count** is a source column that is 1 for every row (△ INFERRED, the standard IBM HR Analytics convention; the values themselves are inside the binary extract so this cannot be confirmed from the workbook. ✗ NOT VERIFIABLE here).
- **Dependencies on other calculated fields:** depends on **Attrition Count**.
- **Aggregation on the shelf:** **derivation='User'**, giving **[usr:Calculation_231935388807876609:qk]**, meaning Tableau treats the field's own aggregate definition as the aggregation rather than wrapping it.
- **Number formatting: none specified.** There is no **number-format** attribute anywhere in this workbook, so this ratio renders with Tableau's default number formatting rather than as a percentage. That is a real, verifiable gap.

Calculated Field: Active Employees

- **Exact Tableau formula:** `SUM([Employee Count]) - SUM([Calculation_231935388807032832])` (written in the Tableau UI as `SUM([Employee Count]) - SUM([Attrition Count])`) **Note the spacing exactly as stored: no space before the minus, one space after.**
- **Internal name:** `[Calculation_231935388808294402]`
- **Data type:** integer
- **Role / type:** measure, quantitative
- **Referenced fields:** `[Employee Count]`, `[Attrition Count]` (calculated)
- **Used in worksheet(s):** KPI only
- **Used in KPI(s):** the Active Employees tile
- **Purpose:** headcount remaining after attrition.
- **Plain English explanation:** total employees minus leavers.
- **Business logic:** treats the population as a closed set where every employee is either active or a leaver. There is no hire date or termination date in the dataset, so this is a snapshot figure, not a point-in-time headcount.
- **Dependencies on other calculated fields:** depends on `Attrition Count`.
- **Aggregation on the shelf:** `derivation='User'`, giving `[usr:Calculation_231935388808294402:qk]`.

Calculated Field: Age (bin)

- **Exact Tableau definition:** `<calculation class='bin' decimals='0' formula='[Age]' peg='0' size-parameter='[Parameters].[Age Parameter]' />`
- **Data type:** integer
- **Role / type:** dimension, ordinal, `aggregation='None'`
- **Referenced fields:** `[Age]`, `[Parameters].[Age Parameter]`
- **Used in worksheet(s):** No. of Employee by Age Group (Columns shelf)
- **Purpose:** bucket employee age into user-controllable bands.
- **Plain English explanation:** put each employee into an age bucket whose width is whatever the Bin Size slider is currently set to, anchored at 0.
- **Business logic:** age band granularity is a user decision, not a fixed design decision.
- **Dependencies:** depends on the `[Age Parameter]` parameter.
- **This is a parameter-driven calculation**, the only one in either workbook.

Calculated Field: `Number of Records`

- **Exact Tableau formula:** `1`
- **Data type:** integer, measure
- **Attribute:** `user:auto-column='numrec'`, that is, auto-generated by Tableau, not hand-created
- **Used in worksheet(s):** none
- **Purpose:** legacy Tableau auto-field. Present but unused.

Calculated Field: `min(1)`

- **Exact Tableau formula:** `min(1)`
- **Internal name:** `[Calculation_1302666200262176769]`
- **Data type:** integer, measure
- **Scope:** ad-hoc inside `Attrition Rate by Gender for Different Age Group`
- **Used in worksheet(s):** that one worksheet
- **Purpose:** axis placeholder for the dual-axis donut.
- **Exact shelf expression:** the Rows shelf is literally `([federated.0f0h03plit3gvo176rhlnly1tx4v].
[usr:Calculation_1302666200262176769:qk] + [federated.0f0h03plit3gvo176rhlnly1tx4v].
[usr:Calculation_1302666200262176769:qk])`
- **Plain English:** a measure that always returns 1. Two copies on Rows create a dual axis so a second, smaller pie can be overlaid to form a donut hole.
- **Note on technique:** Dashboard 1 uses a constant `0` for the same job, Dashboard 2 uses `min(1)`. Both are standard variants of the same trick.

6.1 Dependency chain

```
[Attrition] (source column, string 'Yes'/'No')
  |
  v
Attrition Count = IF [Attrition] = 'Yes' THEN 1 ELSE 0 END
  |
  +-----+
  |               |
  v               v
Attrition Rate   Active Employees
= SUM([Attrition Count]) / SUM([Employee Count])
  |               = SUM([Employee Count]) - SUM([Attrition Count])
  v               |
  v               v
KPI tile 3       KPI tile 4

[Age] + [Age Parameter] "Bin Size" (2.0 to 10.0, step 1.0, current 3.0)
  |
  v
Age (bin) --> Columns shelf of "No. of Employee by Age Group"
```

6.2 Calculation type census for Dashboard 2

Calculation type	Present?	Detail
LOD expressions (FIXED , INCLUDE , EXCLUDE)	No. No { appears in any formula.	
Table calculations	Yes. PctTotal on SUM([Attrition Count]) , giving [pcto:sum:Calculation_231935388807032832:qk]	
Window calculations	No.	
Running totals	No.	
Percent-of-total	Yes , on Department wise Attrition and Attrition Rate by Gender for Different Age Group . The tooltip labels it % of Total Attrition Count along Table (Across) .	
Rank calculations	No.	
Date calculations	No. There is no date field.	
Conditional calculations / IF-THEN-ELSE	Yes , exactly one: Attrition Count .	
CASE statements	No.	
Aggregate calculations	Yes , two: Attrition Rate and Active Employees , both aggregate-over-aggregate.	
Parameters used inside calculations	Yes , one: [Age Parameter] as the size-parameter of [Age (bin)] .	
Aggregations used on shelves	SUM , AVG , User (the field's own aggregate definition), None (discrete dimension)	

7. PARAMETERS

Parameter	Data Type	Current Value	Allowed Values	Used By	Purpose
Bin Size (internal name [Age Parameter])	real (float)	3.0	param-domain-type='range', min='2.0', max='10.0', granularity='1.0' (so effectively 2, 3, 4, 5, 6, 7, 8, 9, 10)	[Age (bin)] as its size-parameter; exposed on the dashboard as a slider control (zone id 13) titled Age Size	Let the viewer change the width of the age buckets at view time

✓ DIRECTLY VERIFIED for every cell.

How the parameter affects the dashboard

The parameter is bound to the size-parameter attribute of the Age (bin) calculation. It does **not** appear in any other formula, on any shelf, in any filter or in any action. Its effect is therefore surgical and single-target:

1. The user drags the Age Size slider on the dashboard between 2 and 10 in steps of 1.
2. Tableau recomputes [Age (bin)] with the new bin width.
3. The No. of Employee by Age Group worksheet redraws with wider or narrower bars. At Bin Size = 2 the histogram is fine-grained; at 10 it collapses into a handful of decade-wide bands.
4. **Nothing else on the dashboard changes.** The KPI strip, the department pie, the job satisfaction table, the education field bars and the age band donuts are all unaffected, because none of them reference [Age (bin)] .

There is a second-order effect worth naming: because No. of Employee by Age Group is the source of filter action [Action2] , which passes [Age (bin)] to the rest of the dashboard, changing the bin size changes **what a click on that chart filters to**. A click at Bin Size = 3 filters a 3-year band; the same click at Bin Size = 10 filters a 10-year band. △ INFERRED from the action and bin definitions, which are both ✓ VERIFIED.

Parameter control formatting

Property	Value
Control zone	id 13, type-v2='paramctrl', mode='slider'
Position	x=55190 y=28000 w=9810 h=5444, that is, overlaid on the upper right of the No. of Employee by Age Group area
Displayed title	Age Size
Title colour	#f1ce63 in Tableau Semibold (dashboard-level parameter-ctrl-title style rule)
Value text colour	#ffffff (dashboard-level parameter-ctrl style rule)

Naming inconsistency worth knowing: the parameter's internal name is `Age Parameter`, its caption is `Bin Size`, and the control on the dashboard displays `Age Size`. Three different names for the same object. All three are ✓ VERIFIED in the XML.

8. FILTERS

8.1 Filter inventory

Filter	Field	Type	Values / Condition	Scope	Effect
Education (global)	<code>[Education]</code> as <code>[none:Education:nk]</code>	Categorical, <code>function='level-members'</code> , <code>ui-enumeration='all'</code>	All members selected	<shared-view> filter , that is, "Apply to all worksheets using this data source". Exposed on the dashboard as a dropdown (zone id 29, sourced from <code>Department wise Attrition</code>)	Filters all 7 worksheets at once, including the KPI strip
Action filter: Department	<code>[Action (Department)]</code>	Categorical, <code>level-members</code> , all, <code>ui-action-filter='[Action1]'</code>	All	Auto-generated by <code>[Action1]</code>	Applied to <code>Attrition Rate by Gender for Different Age Group</code> , <code>Attrition by Gender</code> , <code>Education Field wise Attrition</code> , <code>Job Satisfaction Rating</code> , <code>KPI</code> , <code>No. of Employee by Age Group</code>
Action filter: Department1 (with saved state)	<code>[Action (Department1)]</code>	Categorical, <code>function='member'</code> , <code>member='"R&D"'</code> , <code>ui-enumeration='inclusive'</code> , <code>ui-action-filter='[Action1]'</code>	Only R&D	Applied to the same 6 worksheets	See the important note below.
Action filter: Age (bin)	<code>[Action (Age (bin))]</code>	Categorical, <code>level-members</code> , all, <code>ui-action-filter='[Action2]'</code>	All	Auto-generated by <code>[Action2]</code>	Applied to <code>Attrition Rate by Gender for Different Age Group</code> , <code>Attrition by Gender</code> , <code>Department wise Attrition</code> , <code>Education Field wise Attrition</code> , <code>Job Satisfaction Rating</code> , <code>KPI</code>
Action filter: Education Field	<code>[Action (Education Field)]</code>	Categorical, <code>level-members</code> , all, <code>ui-action-filter='[Action5]'</code>	All	Auto-generated by <code>[Action5]</code>	Applied to <code>Attrition Rate by Gender for Different Age Group</code> , <code>Attrition by Gender</code> , <code>Department wise Attrition</code> , <code>Job Satisfaction Rating</code> , <code>KPI</code> , <code>No. of Employee by Age Group</code>
Measure Names filter	<code>[:Measure Names]</code>	Categorical, <code>function='union'</code> , <code>op='manual'</code> , 5 explicit members	See 8.3	Worksheet-only: <code>KPI</code>	Controls which 5 measures appear as KPI tiles and in what order

8.2 The persisted `R&D` selection (important)

The workbook was saved while a mark representing `Department = "R&D"` was selected on the `Department wise Attrition` chart. Tableau persisted that selection as a member-level action filter:

```
<filter class='categorical' column='[federated.0f0h03plit3gvol76rhlnly1tx4v].[Action (Department1)]'>
  <groupfilter function='member' level='[Department]' member='R&D'
    user:ui-action-filter='[Action1]'
    user:ui-domain='database'
    user:ui-enumeration='inclusive'
    user:ui-marker='enumerate' />
</filter>
```

✓ DIRECTLY VERIFIED. It appears in six worksheets: **Attrition Rate by Gender for Different Age Group**, **Attrition by Gender**, **Education Field wise Attrition**, **Job Satisfaction Rating**, **KPI**, **No. of Employee by Age Group**. It does **not** appear in **Department wise Attrition**, which is the source of the action and so excludes itself.

What this means in practice (△ INFERRED, but a direct consequence of the verified XML): the dashboard as saved carries a live **R&D** filter state on six of its seven views. Depending on how it is opened and whether the action's **auto-clear** fires, a viewer may land on a dashboard that is already filtered to the R&D department rather than showing the full organisation. If you are demoing this, check the published view and be ready to explain it. If it does open filtered, the fix is to deselect the mark and re-save.

8.3 The **KPI** worksheet Measure Names filter

`<groupfilter function='union' user:op='manual'>` containing five explicit **member** entries, in this stored order:

Order	Member	Rendered as
1	<code>[sum:Employee Count:qk]</code>	<code>SUM(Employee Count)</code>
2	<code>[sum:Calculation_231935388807032832:qk]</code>	<code>SUM(Attrition Count)</code>
3	<code>[usr:Calculation_231935388807876609:qk]</code>	<code>Attrition Rate</code>
4	<code>[usr:Calculation_231935388808294402:qk]</code>	<code>Active Employees</code>
5	<code>[avg:Age:qk]</code>	<code>AVG(Age)</code>

A separate `<manual-sort column='[:Measure Names]' direction='ASC'>` carries a dictionary listing only the **first four** members. `[avg:Age:qk]` is not in the sort dictionary, which in Tableau means it sorts after the explicitly ordered members. ✓ VERIFIED.

8.4 Filter types NOT present

Filter type	Present?
Extract filter	No
Context filter	No

Filter type	Present?
Measure filter	No. Every filter is categorical on a dimension or on Measure Names.
Relative date filter	No. There is no date field.
Range / quantitative filter	No
Top N filter	No
Parameter-based filter	No. The parameter drives a bin size, not a filter.
Condition or formula filter	No

8.5 Filter scope map

Worksheet	Education (global)	Action(Department)	Action(Department1) = R&D	Action(Age (bin))	Action(Education Field)	Measure Names
KPI	Yes	Yes	Yes	Yes	Yes	Yes
Attrition by Gender	Yes	Yes	Yes	Yes	Yes	no
Department wise Attrition	Yes	no (source of Action1)	no	Yes	Yes	no
No. of Employee by Age Group	Yes	Yes	Yes	no (source of Action2)	Yes	no
Job Satisfaction Rating	Yes	Yes	Yes	Yes	Yes	no
Education Field wise Attrition	Yes	Yes	Yes	Yes	no (source of Action5)	no
Attrition Rate by Gender for Different Age Group	Yes	Yes	Yes	Yes	Yes	no

Design observation (△ inferred): the Education filter being scoped at the data source level rather than per-worksheet is the technically correct choice for a global control, and it is a meaningfully better pattern than Dashboard 1's filter group, which left the KPI cards out of scope. Here the KPI strip does respond to the global filter.

9. KPI IDENTIFICATION

All five KPIs live in **one** worksheet, **KPI**, built with Measure Names and Measure Values rather than as five separate worksheets. That is the structural difference from Dashboard 1.

Shared context for all five:

Attribute	Value
Worksheet	KPI

Attribute	Value
Columns shelf	<code>[:Measure Names]</code>
Rows shelf	empty
Marks type	Automatic (text)
Marks > Text	<code>[Multiple Values]</code> , that is, Measure Values
Custom label	<code><[Multiple Values]></code>
Custom tooltip	<code><[:Measure Names]>: <[Multiple Values]></code> in bold, measure name in black, value in <code>#2f5597</code> at 14pt
Dashboard zone	id 5, x=1139 y=10667 w=83924 h=15111, <code>show-title='false'</code> . Spans roughly 84 percent of the canvas width in the upper strip.
Filters affecting all five	Education global filter, plus the four action filters including the persisted <code>R&D</code> state
Parameters affecting them	None. <code>Bin Size</code> does not touch the KPI strip.
Time period	Not applicable, no date field
Number formatting	None specified. Defaults apply.

KPI 1: Employee Count

Attribute	Value
Exact displayed name	<code>Employee Count</code> (the Measure Names label for <code>[sum:Employee Count:qk]</code>)
Field behind it	<code>[Employee Count]</code> , source column 19
Exact expression	<code>SUM([Employee Count])</code>
Aggregation	<code>SUM</code>
Numerator / Denominator	Not a ratio
Unit	Count of employees
Business meaning	Total headcount in the filtered population
Why useful	The denominator context for every other figure on the dashboard
What a user learns	The size of the population currently in scope after the Education filter and any cross-filter clicks
Dependency chain	Direct source column, no calculated field involved

KPI 2: Attrition Count

Attribute	Value
Exact displayed name	<code>Attrition Count</code>
Field behind it	<code>[Calculation_231935388807032832]</code>
Exact formula	<code>IF [Attrition] = 'Yes' THEN 1 ELSE 0 END</code> , aggregated as <code>SUM</code>
Aggregation	<code>SUM</code>

Attribute	Value
Numerator	Employees where <code>Attrition = 'Yes'</code>
Denominator	Not applicable
Unit	Count of employees
Business meaning	Number of leavers in the filtered population
Why useful	The headline attrition figure in absolute terms
Dependency chain	<code>[Attrition]</code> (source string) → <code>Attrition Count</code>

KPI 3: Attrition Rate

Attribute	Value
Exact displayed name	<code>Attrition Rate</code>
Field behind it	<code>[Calculation_231935388807876609]</code>
Exact formula	<code>SUM([Calculation_231935388807032832])/SUM([Employee Count])</code>
Aggregation	<code>derivation='User'</code> , the field's own aggregate definition
Numerator	<code>SUM([Attrition Count])</code>
Denominator	<code>SUM([Employee Count])</code>
Unit	A proportion. Not formatted as a percentage , because no <code>number-format</code> is set anywhere in the workbook, so it will render as a decimal unless Tableau's default inference formats it otherwise. ✓ VERIFIED that no format is stored.
Business meaning	Share of the workforce that left
Why useful	Counts are not comparable across departments of different sizes. A rate is.
What a user learns	Whether attrition is high relative to headcount, and how that changes as they cross-filter by department, age band or education field
Dependency chain	<code>[Attrition]</code> → <code>Attrition Count</code> → <code>Attrition Rate</code> . This is the only two-level calculation dependency in either workbook.

KPI 4: Active Employees

Attribute	Value
Exact displayed name	<code>Active Employees</code>
Field behind it	<code>[Calculation_231935388808294402]</code>
Exact formula	<code>SUM([Employee Count]) - SUM([Calculation_231935388807032832])</code>
Aggregation	<code>derivation='User'</code>

Attribute	Value
Numerator / Denominator	Not a ratio, it is a difference
Unit	Count of employees
Business meaning	Headcount remaining after attrition
Why useful	Gives the retained population directly rather than making the reader subtract
Dependency chain	[Attrition] → Attrition Count → Active Employees
Note	Employee Count and Active Employees are shown side by side, so the pair implicitly communicates the attrition gap

KPI 5: Average Age

Attribute	Value
Exact displayed name	the Measure Names label for [avg:Age:qk] , that is AVG(Age)
Field behind it	[Age] , source column 14
Exact expression	AVG([Age])
Aggregation	AVG
Unit	Years
Business meaning	Mean age of the filtered population
Why useful	Frames the age band views below and flags whether a cross-filter has selected an unusually young or old subgroup
Dependency chain	Direct source column
Sort note	This is the one measure not listed in the Measure Names manual-sort dictionary, so it sorts after the other four

KPIs that are NOT present

There is **no target**, **no benchmark**, **no prior-period comparison**, **no variance** and **no trend indicator**. There is also no retention rate (the complement of attrition rate). ✓ VERIFIED by absence.

10. WORKSHEET / SHEET-BY-SHEET ANALYSIS

Seven worksheets exist and **all seven are placed on the dashboard**. None is orphaned.

Shared formatting:

Property	Value
Worksheet title	Six of the seven use the dynamic token <code><Sheet Name></code> in <code>#f1ce63</code> (gold) <code>Tableau Semibold</code> , size 12 (size 11 and <code>fontalignment='1'</code> for <code>Attrition by Gender</code>). The <code>KPI</code> sheet has no title, and its dashboard zone sets <code>show-title='false'</code> .
Tooltip	All seven have a <code><customized-tooltip></code> . The house style is a bold black label, then the value in bold <code>#2f5597</code> (dark blue) at 14pt.
Number format	None specified anywhere.

10.1 KPI

Property	Value
Purpose	Present five headline HR figures in one strip
Visualization type	Text / BAN row driven by Measure Names and Measure Values
Data source	<code>HR data</code> (<code>HR Data</code>)
Columns shelf	<code>[Measure Names]</code>
Rows shelf	empty
Marks type	<code>Automatic</code> (renders as Text)
Marks > Text	<code>[Multiple Values]</code> (Measure Values)
Marks > Color / Size / Detail	none
Custom label	<code><[Multiple Values]></code>
Custom tooltip	<code><[Measure Names]>: then <[Multiple Values]></code>
Filters	Measure Names union of 5 members; Education global; 4 action filters including the <code>R&D</code> state
Pages	none
Parameters	none
Sets / Groups	the 4 hidden action groups
Calculated fields used	<code>Attrition Count</code> , <code>Attrition Rate</code> , <code>Active Employees</code>
Table calculations	none on this sheet
Sorting	<code><manual-sort column='[Measure Names]' direction='ASC'></code> with an explicit dictionary of 4 members
Reference lines / trend lines / forecasting / analytics	none
Dual axis	no
Measure Names / Measure Values	Yes, this worksheet is built on them.

Plain English. Measure Names is placed on Columns, which creates one column per selected measure, and Measure Values goes on Text, which prints the corresponding number in each column. A filter on Measure Names restricts the set to exactly five, and a manual sort pins their left-to-right order. The result is a single worksheet that behaves like five KPI tiles. This is a

more economical construction than Dashboard 1's approach of building five separate worksheets, and it means all five tiles share one set of filters automatically.

10.2 Attrition by Gender

Property	Value
Purpose	Compare absolute attrition between genders
Visualization type	Lollipop chart , built as a dual axis of Bar plus Circle
Rows shelf	[none:Gender:nk] (discrete)
Columns shelf	([sum:Calculation_231935388807032832:qk] + [sum:Calculation_231935388807032832:qk]), that is SUM([Attrition Count]) placed twice
Marks type	Pane 0 Automatic , Pane 1 Bar , Pane 2 Circle (x-index='1' on pane 2)
Marks > Color	[none:Gender:nk] on all panes. No palette override is stored, so Tableau's default categorical palette applies.
Marks > Label	SUM([Attrition Count]) in bold white (fontcolor='#ffffff') on all three panes
Marks > Size / Detail	none
Custom tooltip	Gender: then <Gender>, Attrition Count: then <SUM(Attrition Count)>
Filters	Education global; Action(Department), Action(Department1)=R&D, Action(Age (bin)), Action(Education Field)
Sorting	none
Reference lines / analytics	none
Dual axis	Yes. Two identical SUM([Attrition Count]) on Columns, second pane carries x-index='1'
Synchronized axis	Both axes carry the identical expression so they coincide. No separate synchronize flag is stored. △ INFERRED
Measure Names / Values	not used
Dashboard zone	id 6, x=85127 y=10333 w=13671 h=15444, sitting to the right of the KPI strip

Plain English. Gender goes on Rows, so there is one row per gender. Attrition Count goes on Columns twice, creating two overlapping horizontal axes. The first draws a bar from zero out to the value, the second draws a circle at the value. Overlaying them gives the lollipop form: a thin stem with a dot at the end. Both are coloured by gender and labelled in white bold with the count.

10.3 Department wise Attrition

Property	Value
Purpose	Show how attrition is distributed across departments
Visualization type	Pie chart (single pane, so a pie rather than a donut)
Rows shelf	empty
Columns shelf	empty

Property	Value
Marks type	Pie
Marks > Color	[none:Department:nk] . No palette override stored, so Tableau's default categorical palette applies.
Marks > Angle (wedge-size)	[sum:Calculation_231935388807032832:qk] , that is SUM([Attrition Count])
Marks > Label	Two fields: [pcto:sum:Calculation_231935388807032832:qk] and [sum:Calculation_231935388807032832:qk]
Custom label	<SUM(Attrition Count)> then (<% of Total Attrition Count>), both bold white in Tableau Medium
Custom tooltip	Department:, % of Total Attrition Count along Table (Across):, Attrition Count:
Table calculation	PctTotal , that is Percent of Total. The tooltip confirms the direction as along Table (Across) .
Filters	Education global; Action(Age (bin)); Action(Education Field). Not filtered by its own Department actions.
Sorting	none
Reference lines / analytics	none
Dual axis	no
Role in interactivity	Source worksheet of filter action [Action1] , which filters the rest of the dashboard on Department
Dashboard zone	id 10 , x=1266 y=27111 w=28544 h=40556. Its Department colour legend is a separate zone (id 11) at x=1456 y=57556.

Plain English. An empty viz with Department on Colour and Attrition Count on Angle. Each department becomes a wedge whose size is its share of total leavers, labelled with both the raw count and the percentage. Clicking a wedge fires **Action1** and filters the other six views to that department.

10.4 No. of Employee by Age Group

Property	Value
Purpose	Show the age distribution of headcount, with user-controllable granularity
Visualization type	Bar chart / histogram
Columns shelf	[none:Age (bin):ok] , the parameter-driven bin, used as a discrete ordinal dimension
Rows shelf	[sum:Employee Count:qk] , that is SUM([Employee Count])
Marks type	Automatic . With a continuous measure on Rows and a discrete bin on Columns this resolves to Bar. △ INFERRED from Tableau's mark rules; the XML stores Automatic .
Marks > Color	[sum:Employee Count:qk] with palette='purple_10_0' , type='interpolated' , that is a continuous purple sequential gradient
Marks > Label	[sum:Employee Count:qk] at fontsize='8'
Marks > Size / Detail	none
Custom tooltip	Age (bin): then <Age (bin)> , Employee Count: then <SUM(Employee Count)>
Filters	Education global; Action(Department); Action(Department1)=R&D; Action(Education Field). Not filtered by its own Age (bin) action.
Parameters	[Age Parameter] via the bin. The slider control sits over this chart.
Row banding	band-color='#00000000' (fully transparent), band-size='1' , so banding is effectively switched off

Property	Value
Sorting	none
Reference lines / analytics	none
Dual axis	no
Role in interactivity	Source worksheet of filter action <code>[Action2]</code> , which filters on <code>Age (bin)</code>
Dashboard zone	id <code>12</code> , x=30633 y=27000 w=34367 h=40556, with its colour legend (id <code>14</code>) and the <code>Age Size</code> slider (id <code>13</code>) overlaid

Plain English. Each bar is an age bucket, its height is the headcount in that bucket, and its fill is a purple shade that also encodes headcount, so taller bars are darker. The bucket width is not fixed: it is whatever the `Age Size` slider says, so the same chart can be read at 2-year or 10-year granularity. Clicking a bar filters the rest of the dashboard to that age band.

10.5 Job Satisfaction Rating

Property	Value
Purpose	Cross-tabulate job role against satisfaction score
Visualization type	Highlight table (heatmap of squares with numbers)
Rows shelf	<code>[none:Job Role:nk]</code>
Columns shelf	<code>[none:Job Satisfaction:ok]</code> (the ordinal, role-converted field)
Marks type	Square
Marks > Color	<code>[sum:Employee Count:qk]</code> with <code>palette='blue_10_0'</code> , <code>type='interpolated'</code> , that is a continuous blue sequential gradient
Marks > Label	<code>[sum:Employee Count:qk]</code>
Marks > Size / Detail	none
Custom tooltip	<code>Job Role: , Job Satisfaction: , Employee Count:</code>
Filters	Education global; Action(Department); Action(Department1)=R&D; Action(Age (bin)); Action(Education Field)
Row banding	<code>band-color='#00000000'</code> on two rules, <code>band-size='1'</code> , so banding is off
Sorting	none
Reference lines / analytics	none
Dual axis	no
Role in interactivity	Source worksheet of filter action <code>[Action3]</code>
Dashboard zone	id <code>23</code> , x=65759 y=27222 w=33228 h=40444. No colour legend is exposed for this chart on the dashboard.

Plain English. Job roles run down the rows and the satisfaction rating runs across the columns. Each cell is a square whose blue intensity and printed number both show how many employees are in that role-and-rating combination. Reading a row tells you how satisfaction is distributed within a role; reading a column tells you which roles cluster at a given rating.

Observation (△ inferred): the chart is titled `Job Satisfaction Rating` and is measured in `Employee Count`, not in `Attrition Count`. It is a headcount distribution view, not an attrition view, which makes it the one chart on the dashboard that answers a different question from the rest.

10.6 Education Field wise Attrition

Property	Value
Purpose	Rank education fields by attrition volume
Visualization type	Horizontal bar chart
Rows shelf	<code>[none:Education Field:nk]</code>
Columns shelf	<code>[sum:Calculation_231935388807032832:qk]</code> , that is <code>SUM([Attrition Count])</code>
Marks type	<code>Automatic</code> , resolving to Bar. △ INFERRED
Marks > Color / Size / Label / Detail	none . This is the plainest chart on the dashboard.
Custom tooltip	<code>Education Field: then <Education Field>, Attrition Count: then <SUM(Attrition Count)></code>
Filters	Education global; Action(Department); Action(Department1)=R&D; Action(Age (bin)). Not filtered by its own Education Field action.
Sorting	none defined . Bars will appear in the data source's default order, not sorted by value. ✓ VERIFIED by absence of any <code><sort></code> or <code><manual-sort></code> on this sheet.
Reference lines / analytics	none
Dual axis	no
Role in interactivity	Source worksheet of filter action <code>[Action5]</code>
Dashboard zone	id <code>24</code> , x=1139 y=68778 w=27278 h=29111

Plain English. One horizontal bar per education field, length equal to the number of leavers from that field. No colour, no label, tooltip only. Clicking a bar filters the dashboard to that education field.

10.7 Attrition Rate by Gender for Different Age Group

Property	Value
Purpose	Break attrition down by gender within each age band
Visualization type	A row of donut charts , one per age band, built as a dual-axis pie
Columns shelf	<code>[none:CF_age band:nk]</code>
Rows shelf	<code>([usr:Calculation_1302666200262176769:qk] + [usr:Calculation_1302666200262176769:qk])</code> , that is <code>min(1)</code> placed twice
Marks type	<code>Pie</code> on all three panes, pane 2 carries <code>y-index='1'</code>
Marks > Color	Pane 1: <code>[none:Gender:nk]</code> . Pane 2: no colour field.
Marks > Angle (wedge-size)	Pane 1: <code>[sum:Calculation_231935388807032832:qk]</code> , that is <code>SUM([Attrition Count])</code>

Property	Value
Marks > Label	Pane 1: [pcto:sum:...] and [sum:...]. Pane 2: [sum:...] at fontsize='15'.
Custom label (pane 1)	<SUM(Attrition Count)> (<% of Total Attrition Count>)
Custom label (pane 2)	<SUM(Attrition Count)> at 15pt, that is the total count printed in the centre of the donut hole
Custom tooltip (pane 1)	CF_age band:, Gender:, % of Total Attrition Count along Table (Across):, Attrition Count:
Custom tooltip (pane 2)	CF_age band:, Attrition Count:
Table calculation	PctTotal, direction along Table (Across) per the tooltip
Filters	Education global; all four action filters including R&D
Sorting	<manual-sort column='[none:CF_age band:nk]' direction='ASC'> with an explicit dictionary: "Under 25", "25 - 34", "35 - 44", "45 - 54", "Over 55". This forces logical age order instead of alphabetical.
Column header formatting	CF_age band header text #ffffff in Tableau Semibold; row axes hidden (display set to false for both classes)
Reference lines / analytics	none
Dual axis	Yes , via two min(1) on Rows
Role in interactivity	Source worksheet of filter action [Action4]
Dashboard zone	id 25, x=29177 y=68889 w=69494 h=28333, the widest chart on the dashboard. Its Gender legend is zone 26.

Plain English. Age band goes on Columns, so there is one cell per band, ordered Under 25 through Over 55 by an explicit manual sort. In each cell, min(1) on Rows twice gives a dual axis. The first pie is coloured by gender with wedge angle equal to that gender's leaver count, labelled with count and percent of total across the row. The second pie is drawn on the secondary axis carrying only a large 15pt label, which places the age band's total attrition count in the middle of the donut. The row axes are hidden so only the donuts and the age band headers show.

A precise and important naming note. The worksheet is called Attrition Rate by Gender for Different Age Group, but the measure actually on the Angle shelf is SUM([Attrition Count]), **not** Attrition Rate. The Attrition Rate calculated field appears **only** on the KPI worksheet. The percentages shown in the labels are % of Total Attrition Count computed by a table calculation, which is a share of leavers across the row, not an attrition rate against headcount. ✓ DIRECTLY VERIFIED from the shelves and the tooltip text. This is a genuine discrepancy between the chart title and the chart content, and it is exactly the kind of thing a sharp interviewer will probe. It is better to name it yourself than to be caught by it.

11. VISUALIZATION-BY-VISUALIZATION BREAKDOWN

#	Visual	What it shows	Fields driving it	Dimensions compared	Measure analyzed	Aggregation	Filters affecting it	Calculations affecting it	Business question answered	Why appropriate
1	KPI strip (5 tiles in one sheet)	Headcount, leavers, attrition rate, active headcount, mean age	Employee Count, Attrition Count, Attrition Rate, Active Employees, Age, Measure Names/Values	none	five measures	SUM, SUM, User, User, AVG	Education global + 4 action filters	Attrition Count → Attrition Rate and Active Employees	"What is the current state of the workforce and its attrition?"	Measure Names/Values gives five synchronised figures in one sheet with one filter scope
2	Attrition by Gender (lollipop)	Leaver counts for each gender	Gender, Attrition Count	Gender	attrition count	SUM	Education global + 4 action filters	Attrition Count	"Do men and women leave at different volumes?"	A lollipop reads cleanly with only two categories in a narrow zone, and carries less ink than a full bar
3	Department wise Attrition (pie)	Share of total leavers by department	Department, Attrition Count	Department	attrition count and share	SUM + PctTotal	Education global, Action(Age (bin)), Action(Education Field)	Attrition Count, percent of total	"Which departments account for most of our attrition?"	A pie is defensible here because the question is explicitly about share of a whole, and the labels carry the raw counts too
4	No. of Employee by Age Group (bar)	Headcount by age bucket at user-chosen granularity	Age (bin), Employee Count, [Age Parameter]	Age buckets	headcount	SUM	Education global, Action(Department), Action(Department1)=R&D, Action(Education Field)	Age (bin) driven by Bin Size	"What is our age profile, and does it change shape at a different band width?"	A histogram is the standard form; making the bin width a parameter lets the viewer test whether a pattern is real or an artefact of bucketing
5	Job Satisfaction Rating (highlight table)	Headcount for every job role and satisfaction score combination	Job Role, Job Satisfaction, Employee Count	Job Role x Job Satisfaction	headcount	SUM	Education global + 4 action filters	none	"Which roles cluster at low satisfaction?"	A highlight table handles a two-dimensional categorical cross-tab better than any chart, and printing the value keeps it readable as a table
6	Education Field wise Attrition (bar)	Leaver counts by education field	Education Field, Attrition Count	Education Field	attrition count	SUM	Education global, Action(Department), Action(Department1)=R&D, Action(Age (bin))	Attrition Count	"Do leavers come disproportionately from particular academic backgrounds?"	Horizontal bars handle long category labels well
7	Attrition Rate by Gender for Different Age Group (donut row)	Gender split of leavers within each age band, plus the band total	CF age band, Gender, Attrition Count	Age band x Gender	attrition count and share	SUM + PctTotal	Education global + 4 action filters	Attrition Count, percent of total, min(1) axis	"Within each age band, is attrition concentrated in one gender?"	Small multiples of donuts let you compare the same two-part split across five bands side by side; the hole carries the band total so the reader gets absolute and relative at once

12. DASHBOARD LAYOUT

Property	Value
Dashboard name	HR Dashboard (trailing space)
Size	1580 x 900 px, <code>sizing-mode='fixed'</code>
Layout style	Floating over a full-canvas background image. Only two objects sit inside the tiled container; the other twelve are floating siblings positioned absolutely.
Background	A PNG image object fills the canvas: <code>Image/HR background.pptx.png</code> , 410,683 bytes, zone id <code>3</code> , <code>type-v2='bitmap'</code> , at x=506 y=889 w=98988 h=98222. The filename <code>HR background.pptx.png</code> indicates it was exported from a PowerPoint file. △ INFERRED from the filename; ✓ VERIFIED that the file and reference exist.
Container structure	<code>layout-basic</code> (id <code>4</code> , <code>margin='8'</code>) containing only the bitmap zone (id <code>3</code> , <code>margin='4'</code>). All remaining zones are siblings at the top <code><zones></code> level, that is, floating.
Device layouts	One auto-generated Phone layout, <code>sizing-mode='vscroll'</code> , height 2400 px, <code>auto-generated='true'</code> . No Tablet layout.

Object census

Object type	Count	Zone ids
Worksheets	7	5 (KPI), 6 (Attrition by Gender), 10 (Department wise Attrition), 12 (No. of Employee by Age Group), 23 (Job Satisfaction Rating), 24 (Education Field wise Attrition), 25 (Attrition Rate by Gender...)
Image object	1	3, <code>Image/HR background.pptx.png</code>
Text object	1	7, containing the text <code>HR ANALYTICS DASHBOARD</code>
Colour legends	3	11 (Department, from Department wise Attrition), 14 (SUM(Employee Count), from No. of Employee by Age Group), 26 (Gender, from Attrition Rate by Gender...)
Filter control	1	29, <code>mode='dropdown'</code> , <code>param='[none:Education:nk]'</code> , sourced from <code>Department wise Attrition</code>
Parameter control	1	13, <code>mode='slider'</code> , <code>param='[Parameters].[Age Parameter]'</code> , titled <code>Age Size</code>
Blank objects	0	
Navigation buttons	0	
Extension / web page objects	0	

Note: no colour legend is exposed for `Job Satisfaction Rating`, even though it has a continuous blue colour encoding. ✓ VERIFIED by absence.

Reconstructed hierarchy

```
HR Dashboard (1580 x 900, FIXED sizing)
├── layout-basic [id 4, margin 8]
│   ├── Image object [id 3] "Image/HR background.pptx.png"
│   │   full canvas: x=506 y=889 w=98988 h=98222
│   └── HEADER BAND (y ~ 2000 to 9500)
│       ├── Text object [id 7] "HR ANALYTICS DASHBOARD"
│       │   x=1203 y=2111 w=65063 h=7222
│       └── Filter: Education (dropdown) [id 29]
│           x=61709 y=2333 w=14620 h=6889
├── KPI BAND (y ~ 10300 to 25800)
│   ├── KPI worksheet [id 5] show-title=false
│   │   x=1139 y=10667 w=83924 h=15111
│   └── Attrition by Gender [id 6]
│       x=85127 y=10333 w=13671 h=15444
├── MIDDLE BAND (y ~ 27000 to 67700)
│   ├── Department wise Attrition [id 10]
│   │   x=1266 y=27111 w=28544 h=40556
│   │   └── Legend: Department [id 11] x=1456 y=57556
│   ├── No. of Employee by Age Group [id 12]
│   │   x=30633 y=27000 w=34367 h=40556
│   │   ├── Legend: SUM(Employee Count) [id 14] x=30886 y=31000
│   │   └── Parameter slider "Age Size" [id 13] x=55190 y=28000
│   └── Job Satisfaction Rating [id 23]
│       x=65759 y=27222 w=33228 h=40444
│       (no legend exposed)
└── BOTTOM BAND (y ~ 68800 to 97900)
    ├── Education Field wise Attrition [id 24]
    │   x=1139 y=68778 w=27278 h=29111
    └── Attrition Rate by Gender for Different Age Group [id 25]
        x=29177 y=68889 w=69494 h=28333
        └── Legend: Gender [id 26] x=70316 y=69889 w=24114 h=3444
```

Zone **x**, **y**, **w**, **h** are in hundred-thousandths of the dashboard dimension. Multiply by 1580/100000 horizontally and 900/100000 vertically for pixels. For example the KPI zone at **x=1139 w=83924** is about 18 px from the left and about 1326 px wide.

Headers and footers: the header band is a genuine text object plus a filter control, unlike Dashboard 1 where the header was implied by worksheet placement. There is no footer object. ✓ VERIFIED.

13. DASHBOARD ACTIONS / INTERACTIVITY

Six filter actions. Every worksheet except **KPI** is the source of exactly one.

Action	Caption	Type	Source worksheet	Target	Trigger	Fields
[Action1]	Filter 1 (generated)	Filter (command='tsc:tsl-filter')	Department wise Attrition	HR Dashboard (whole dashboard)	on-select, auto-clear='true'	special-fields='all'
[Action2]	Filter 2 (generated)	Filter	No. of Employee by Age Group	HR Dashboard	on-select, auto-clear='true'	special-fields='all'
[Action3]	Filter 3 (generated)	Filter	Job Satisfaction Rating	HR Dashboard	on-select, auto-clear='true'	special-fields='all'
[Action4]	Filter 4 (generated)	Filter	Attrition Rate by Gender for Different Age Group	HR Dashboard	on-select, auto-clear='true'	special-fields='all'
[Action5]	Filter 5 (generated)	Filter	Education Field wise Attrition	HR Dashboard	on-select, auto-clear='true'	special-fields='all'
[Action6]	Filter 6 (generated)	Filter	Attrition by Gender	HR Dashboard	on-select, auto-clear='true'	special-fields='all'

✓ DIRECTLY VERIFIED for every cell.

What **special-fields='all'** means

The actions pass **all fields** in the source view rather than a named subset. So clicking a wedge on **Department wise Attrition** passes **Department** to every target view; clicking a square on **Job Satisfaction Rating** passes both **Job Role** and **Job Satisfaction**. This is the "Use all fields" option in the Tableau action dialog, and it is why the auto-generated hidden groups (**Action (Department)** , **Action (Age (bin))** , **Action (Education Field)**) exist.

Result and effect

Every chart is a cross-filter surface. Selecting a mark anywhere filters the other six views, including the KPI strip, and deselecting restores the full view because **auto-clear='true'**. Combined with the global Education dropdown, a user can compose a query like "R&D department, science education field, 35 to 44 age band" purely by clicking, and the KPI strip recomputes headcount, leaver count, attrition rate, active headcount and mean age for that slice.

Interactivity types NOT present

Interactivity	Present?
Highlight actions	No. Contrast with Dashboard 1, which has exactly one highlight action and no filter actions.

Interactivity	Present?
URL actions	No
Parameter actions	No. The parameter is driven by its slider only, not by clicking a mark.
Set actions	No (no sets exist)
Navigation / Go to Sheet actions	No
Sheet swapping	No
Dashboard-to-dashboard navigation	No (only one dashboard)
Drill-down hierarchies	No (no hierarchies defined)
Viz-in-tooltip	No

Interactive controls available to the end user

1. Education dropdown filter, scoped to all seven worksheets.
2. **Age Size** parameter slider (2 to 10, step 1), affecting **No. of Employee by Age Group** and, indirectly, what **[Action2]** filters to.
3. Six click-to-filter actions, one from every chart.
4. Three colour legends, which in Tableau also support highlight-on-click.

14. TOOLTIPS

Every one of the seven worksheets has a custom tooltip. That is the sharpest contrast with Dashboard 1, which has none. ✓ DIRECTLY VERIFIED.

House style, consistent across all of them: the field label in **bold #000000** , then the value in **bold #2f5597** (dark blue) at **14pt**, with **Æ** separators (which is how a line break renders in the raw XML text runs).

Worksheet	Fields in the tooltip	Calculated values?	Custom descriptions?	Viz-in-tooltip?	Conditional?
KPI	<Measure Names>: then <Multiple Values>	Yes, Attrition Rate and Active Employees flow through Measure Values	Yes, the label is hand-typed	No	No
Attrition by Gender	Gender: <Gender>; Attrition Count: <SUM(Attrition Count)>	Yes, Attrition Count	Yes	No	No
Department wise Attrition	Department: <Department>; % of Total Attrition Count along Table (Across): <pcto SUM(Attrition Count)>; Attrition Count: <SUM(Attrition Count)>	Yes, plus a table calculation	Yes	No	No

Worksheet	Fields in the tooltip	Calculated values?	Custom descriptions?	Viz-in-tooltip?	Conditional?
No. of Employee by Age Group	Age (bin): <Age (bin)>; Employee Count: <SUM(Employee Count)>	Yes, the parameter-driven bin	Yes	No	No
Job Satisfaction Rating	Job Role: <Job Role>; Job Satisfaction: <Job Satisfaction>; Employee Count: <SUM(Employee Count)>	No	Yes	No	No
Education Field wise Attrition	Education Field: <Education Field>; Attrition Count: <SUM(Attrition Count)>	Yes, Attrition Count	Yes	No	No
Attrition Rate by Gender for Different Age Group (pane 1)	CF_age band: <CF_age band>; Gender: <Gender>; % of Total Attrition Count along Table (Across): <pcto>; Attrition Count: <SUM(Attrition Count)>	Yes, plus a table calculation	Yes	No	No
Attrition Rate by Gender for Different Age Group (pane 2, the donut hole)	CF_age band: <CF_age band>; Attrition Count: <SUM(Attrition Count)>	Yes	Yes	No	No

Why this matters (△ inferred). The tooltips are where the `Name1` caption artefacts get corrected. The field is captioned `Department1` in the data pane, but the tooltip reads `Department:` . The same holds for `Gender`, `Job Role`, `Education Field` and `Job Satisfaction`. Someone hand-wrote clean labels into every tooltip.

Custom mark labels

Worksheet	Custom label content
Attrition by Gender	<SUM(Attrition Count)> in bold #ffffff, on all three panes
Department wise Attrition	<SUM(Attrition Count)> then (<% of Total Attrition Count>), bold #ffffff in Tableau Medium
No. of Employee by Age Group	<SUM(Employee Count)> at fontsize='8'
KPI	<Multiple Values>
Attrition Rate by Gender... pane 1	<SUM(Attrition Count)> (<% of Total Attrition Count>)
Attrition Rate by Gender... pane 2	<SUM(Attrition Count)> at fontsize='15', the number placed in the donut hole
Job Satisfaction Rating	none (the default label from the Text shelf)
Education Field wise Attrition	none

Worksheet titles

Six worksheets use the dynamic `<Sheet Name>` token so the title always matches the worksheet name:

Worksheet	Title definition	Font
Attrition Rate by Gender for Different Age Group	<Sheet Name>	#f1ce63, Tableau Semibold, 12
Attrition by Gender	<Sheet Name>	#f1ce63, Tableau Semibold, 11, centred
Department wise Attrition	<Sheet Name>	#f1ce63, Tableau Semibold, 12

Worksheet	Title definition	Font
Education Field wise Attrition	<Sheet Name>	#f1ce63, Tableau Semibold, 12
Job Satisfaction Rating	<Sheet Name>	#f1ce63, Tableau Semibold, 12
No. of Employee by Age Group	<Sheet Name>	#f1ce63, Tableau Semibold, 12
KPI	no custom title; hidden on the dashboard	n/a

Using <Sheet Name> rather than typed text means renaming a worksheet automatically renames its dashboard title. That is a small but real maintainability choice. Δ INFERRED as intent; $\checkmark$ VERIFIED as fact.

15. BUSINESS LOGIC

What business problem was identified (Δ INFERRED). Attrition is expensive and uneven, and an HR team needs to know where it concentrates before it can act. The dashboard is built to answer "where", not "why" or "who next": it is a descriptive, cross-filterable profile, not a predictive model and not a driver analysis.

What data was needed ($\checkmark$). One employee-level file at one row per employee, 39 columns, containing an attrition flag, headcount, department, education field, education level, gender, job role, job satisfaction score, age, and a pre-computed age band. Twenty-four of the thirty-nine available columns were not used.

How the data was transformed ($\checkmark$). 1. `HR_data.csv` was read with a `textscan` connection and extracted whole to Hyper. 2. `Job Satisfaction` and `Employee Number` were converted from measures to ordinal dimensions. 3. `Attrition Count` was calculated as `IF [Attrition] = 'Yes' THEN 1 ELSE 0 END`, converting a string flag into a summable indicator. 4. `Attrition Rate` was calculated as `SUM([Attrition Count])/SUM([Employee Count])`. 5. `Active Employees` was calculated as `SUM([Employee Count]) - SUM([Attrition Count])`. 6. A `Bin Size` parameter (real, 2.0 to 10.0, step 1.0, default 3.0) was created and bound to an `Age (bin)` binning calculation as its `size-parameter`. 7. A global Education filter was set at data source scope via a `<shared-view>`. 8. `CF_age band`, `CF_attrition label` and `CF_current Employee` arrived already derived from upstream and were consumed as-is.

What KPIs were selected ($\checkmark$). `Employee Count`, `Attrition Count`, `Attrition Rate`, `Active Employees`, `AVG(Age)`. Four counts or differences and one true ratio.

What dimensions were analysed ($\checkmark$). `Department`, `Education Field`, `Education`, `Gender`, `Job Role`, `Job Satisfaction`, `CF age band`, and `Age (bin)`.

What comparisons were made ($\checkmark$). - Attrition volume between genders (lollipop). - Attrition share across departments (pie with percent of total). - Headcount across age buckets at user-chosen granularity (parameterised histogram). - Headcount across the job role by satisfaction matrix (highlight table). - Attrition volume across education fields (bar). -

Gender split of attrition within each of five age bands, with the band total in the hole (donut small multiples with percent of total and a manual age-band sort).

What trends were intended to be identified (△ INFERRED). The design points at concentration rather than trend.

There is no time series and there cannot be one, because the dataset has no date field. The tenure columns (*Years At Company* , *Years Since Last Promotion* , and so on) could have supported a tenure-based view but are not used. Be precise about this in an interview: this dashboard shows *where* attrition sits, not *how it is moving*.

What decisions a business user could make (△ INFERRED). Target retention effort at the department, age band, education field or job role with the worst concentration; decide whether a gender-specific retention issue exists in a particular age band; identify job roles whose satisfaction distribution skews low and pair that with their attrition contribution; and scope a deeper analysis by using the cross-filters to isolate a cohort before requesting record-level data.

16. KEY INSIGHTS THE DASHBOARD IS DESIGNED TO ENABLE

No actual data values were read, so no findings are claimed.

A. Insights explicitly encoded by the dashboard

1. Headcount, leaver count, attrition rate, active headcount and mean age are computed for whatever slice is currently filtered, and recompute on every cross-filter click.
2. Attrition is decomposed by department as a share of total leavers, with both the count and the percent of total on the label.
3. Attrition is decomposed by education field as an absolute count.
4. Attrition is decomposed by gender as an absolute count.
5. Attrition is decomposed by gender **within** each of five age bands, with both the count, the percent of total across the band row, and the band total in the donut hole.
6. Headcount is decomposed by age bucket at a granularity the user chooses between 2 and 10 years.
7. Headcount is cross-tabulated across job role and job satisfaction score, encoded by both colour intensity and printed value.
8. Selecting any mark on any chart re-filters the entire dashboard including the KPI strip.

B. Questions the dashboard enables users to answer

- "What is our attrition rate, and how many people does that represent?"
- "Which department contributes the largest share of our leavers?"

- "Within the 25 to 34 age band, is attrition skewed towards one gender?"
- "If I restrict to a single education level, does the department picture change?"
- "Which job roles have the largest concentration of employees at the lowest satisfaction rating?"
- "Is our age distribution bimodal, or does that disappear when I widen the bins from 3 years to 8?"
- "For the R&D department specifically, what are the headcount, leaver count and attrition rate?" (by clicking the R&D wedge)
- "Does the age profile of the population change when I filter to a particular education field?"

C. Actual findings verifiable from the available data

None can be stated. All values live in the binary `.hyper` extracts and no Hyper reader is available here. No headcount, no attrition rate, no departmental ranking and no satisfaction distribution can be verified from the files supplied. **X NOT AVAILABLE.**

One thing that **is** verifiable and worth noting as a caveat rather than a finding: the saved workbook state carries a `Department = "R&D"` action filter on six of the seven worksheets (section 8.2). Any number read off the dashboard in that state is an R&D number, not an organisation-wide number.

17. TECHNICAL IMPLEMENTATION SUMMARY

DATA SOURCE

HR_data.csv, 39 columns, one row per employee
/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main
(3 further data sources point at variants of the same file and are UNUSED)
3 columns arrive pre-derived from upstream: CF_age band,
CF_attrition label, CF_current Employee

|

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DATA CONNECTION

Tableau textscan (flat file) connector
(an excel-direct connection is also declared but not bound to the relation)
wrapped in federated.0f0h03plit3gvo176rhlnlyltx4v
Extract enabled, count = -1 (all records), no extract filter
4 .hyper files ship in the archive, 3 of them dead weight

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DATA MODEL

Single table. No joins, no relationships, no unions, no blends.

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DATA CLEANING (all inside Tableau except the CF_ columns)

10 field captions (6 are Name1 collision artefacts, not deliberate renames)
Job Satisfaction and Employee Number converted measure -> ordinal dimension
Attrition Count converts a 'Yes'/'No' string into a summable 0/1 flag
Age binned with a PARAMETER-DRIVEN bin width
No aliases, no groups, no sets, no hierarchies

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CALCULATED FIELDS (6 total, no LODs)

Attrition Count = IF [Attrition] = 'Yes' THEN 1 ELSE 0 END
Attrition Rate = SUM([Attrition Count]) / SUM([Employee Count])
Active Employees = SUM([Employee Count]) - SUM([Attrition Count])
Age (bin) = bin of [Age], size-parameter = [Age Parameter]
Number of Records = 1 (auto-generated, unused)
min(1) = axis placeholder for the dual-axis donut
+ built-in table calculation: Percent of Total, along Table (Across)

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PARAMETER (1)

[Age Parameter] "Bin Size", real, range 2.0 to 10.0, step 1.0, current 3.0
Surfaced as a slider titled "Age Size"

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KPIs (5, all in ONE worksheet via Measure Names / Measure Values)

SUM(Employee Count) -> Employee Count
SUM(Attrition Count) -> Attrition Count
Attrition Rate -> Attrition Rate

```

Active Employees      -> Active Employees
AVG(Age)              -> Average Age
|
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WORKSHEETS  (7)
1 Measure Names/Values KPI strip (manual-sorted)
1 dual-axis lollipop      (Bar + Circle)
1 pie with percent of total
1 parameterised bar histogram (purple sequential colour)
1 highlight table        (Square marks, blue sequential colour)
1 horizontal bar
1 dual-axis donut small multiples (manual-sorted age bands,
                                total printed in the hole)
|
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DASHBOARD  (1)
"HR Dashboard " (trailing space), 1580 x 900 FIXED
Floating layout over a full-canvas PNG exported from PowerPoint
7 worksheets + 1 text title + 3 colour legends
+ 1 filter dropdown + 1 parameter slider
Gold (#f1ce63) title system, dark-blue (#2f5597) tooltip values
Auto-generated Phone layout (vscroll, 2400 px)
|
v
FILTERS / ACTIONS
Education filter at DATA SOURCE scope (shared-view) -> all 7 worksheets
6 filter actions, one from every chart, on-select, auto-clear,
    special-fields = all, targeting the whole dashboard
4 auto-generated hidden action groups
A persisted Department = "R&D" selection saved into 6 worksheets
|
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FINAL USER EXPERIENCE
A single fixed-size branded screen. Pick an education level from the
dropdown, drag the age bin slider to change histogram granularity, then
click any wedge, bar, square or donut segment to re-filter the whole
dashboard including the five KPI figures. Every mark carries a hand-written
tooltip. Deselecting restores the full view.

```

18. INTERVIEW EXPLANATION

Use this as a spoken narrative. Everything in it is supported by the workbook.

"This is an HR attrition dashboard I built for myself, to practise metric design and interaction design rather than just chart building. The dataset is a public employee-level set from Kaggle, 39 columns, one row per employee, with an attrition flag, department, education field and level, gender, job role, a job satisfaction score, age and a pre-computed age band.

The problem I set out to solve was that HR teams usually know their overall attrition number and almost nothing about where it concentrates. So I built the dashboard as a cross-filtering surface rather than a static report. Every chart on it is clickable, and clicking anything re-filters everything else including the headline numbers.

The data came in as a single CSV read through Tableau's flat file connector and extracted whole to Hyper. One table, no joins. I will be straight about something visible in the file: there are four data sources in that workbook pointing at variants of the same file, and only one is actually wired up. That is leftover from trying different import formats early on, and if I were tidying it I would delete the three dead ones, because each one carries its own extract and bloats the packaged file.

The calculation layer is where the real design decisions are. The attrition column arrives as the text Yes or No, which you cannot aggregate, so the first thing I built was Attrition Count: `IF Attrition equals Yes THEN 1 ELSE 0 END`. Everything else on the dashboard hangs off that. Attrition Rate is SUM of Attrition Count divided by SUM of Employee Count. I wrote it as an aggregate over an aggregate on purpose, because that way it recalculates correctly at whatever level of detail the view happens to be at. Put it on a KPI tile with no dimensions and it is the organisation-wide rate; click into a department and the same field gives you that department's rate without me having to write a second calculation. Active Employees is SUM of Employee Count minus SUM of Attrition Count, so the strip shows total, leavers, rate and remaining side by side.

The piece I am most pleased with is the parameterised bin. Instead of hard-coding an age bucket width, I created a Bin Size parameter, a float between 2 and 10 stepping by 1, defaulting to 3, and bound it to the bin's size-parameter property. Then I put it on the dashboard as a slider. That means the viewer can decide whether they want a fine-grained age profile or decade bands, and they can check whether a pattern they think they see survives a change of bucketing. It also has a second effect I had to think through: that histogram is a filter source, so changing the bin width changes the width of the band a click filters to.

For the KPI strip I did not build five separate worksheets. I used Measure Names on Columns and Measure Values on Text in a single sheet, filtered Measure Names down to the five I wanted, and pinned their order with a manual sort. One sheet, five tiles, one filter scope, much less to maintain.

The global Education filter is set at data source scope using `apply-to-all-worksheets-using-this-data-source`, rather than picking sheets by hand. That guarantees the KPI strip responds to it, which matters, because a filter that moves the charts but not the headline numbers is worse than no filter at all.

Chart choices: a lollipop for gender attrition, which is a dual axis of a bar and a circle on the same measure, because with two categories in a narrow panel it reads cleaner than a full bar. A pie for department share, which I would normally avoid, but the question there is genuinely about share of a whole and I put the raw counts on the labels too. A highlight

table of job role against satisfaction score with Square marks, because a two-way categorical cross-tab is what a highlight table is for. A parameterised histogram for age. Plain horizontal bars for education field. And the bottom chart is a row of donuts, one per age band, gender-split, with the band's total attrition count printed in the middle of each hole using the second axis of the dual-axis pie. I forced the age bands into logical order with a manual sort dictionary, Under 25 through Over 55, because alphabetical order would have put Over 55 in the middle.

Every worksheet has a hand-written tooltip in a consistent style, bold black label and dark blue value. The worksheet titles use the Sheet Name token rather than typed text, so renaming a sheet renames its dashboard title automatically.

Layout is 1580 by 900 fixed, floating objects over a full-canvas background image I exported from PowerPoint, with a gold title system.

If I am being properly honest about what I would fix, there are four things I can see in my own file. First, the bottom chart is titled Attrition Rate by Gender for Different Age Group, but the measure on it is actually Attrition Count with a percent-of-total table calculation, not the Attrition Rate field. I would either rename the sheet or swap in the rate. Second, I set no number formatting anywhere, so Attrition Rate renders as a decimal rather than a percentage. Third, six of my field captions ended up as Department1, Gender1 and so on, which are Tableau collision artefacts from replacing a data source, not deliberate names. My tooltips paper over that but the legends do not. Fourth, I saved the workbook with an R&D mark still selected, so the saved state carries a department filter on six of seven sheets.

Skills-wise this demonstrates parameter-driven calculations, aggregate calculations and a real dependency chain, Measure Names and Measure Values KPI construction, dual-axis chart building for both lollipops and donuts, table calculations, data source scoped filtering, six-way action-based cross-filtering, manual sort dictionaries, sequential colour encoding, custom tooltip authoring and floating dashboard layout over a designed background."

19. RESUME-READY PROJECT DESCRIPTION

Only verified facts. No invented percentages, record counts or outcomes.

HR Analytics Attrition Dashboard (Tableau)

- Built a 7-worksheet Tableau attrition dashboard over a 39-column employee-level dataset, published to Tableau Public on a fixed 1580x900 canvas with a floating layout over a custom-designed background.
- Authored the metric layer as a three-level calculation chain: converted a Yes / No text attrition flag into a summable indicator with `IF [Attrition] = 'Yes' THEN 1 ELSE 0 END`, then derived `Attrition Rate` as `SUM([Attrition Count])/SUM([Employee Count])` and `Active Employees` as `SUM([Employee Count]) - SUM([Attrition Count])` as aggregate calculations so both recompute correctly at any level of detail.

- Implemented a parameter-driven age histogram by binding a `Bin Size` range parameter (2 to 10, step 1) to the `size-parameter` of an `Age (bin)` calculation and exposing it as a dashboard slider, letting users change bucket granularity at view time.
- Condensed five KPIs into a single worksheet using Measure Names and Measure Values with a manual-sort dictionary, and scoped the Education filter at data source level so the KPI strip and all six charts respond to one control.
- Delivered full cross-filtering with six `on-select` filter actions (one per chart, passing all fields, auto-clear) alongside dual-axis chart construction for a Bar-plus-Circle lollipop and a row of donut small multiples, Percent of Total table calculations, sequential colour encodings, and hand-authored tooltips on all seven worksheets.

PART B

Hands-on evidence brief

Sections 1 to 13 for a hiring manager confirming hands-on experience: context, source inventory, source-to-target mapping, data flow, cleaning rules, data model, business rules and governance, operations and security, requirements to KPIs, stakeholder coordination, outcomes, open questions and a sanitised version with its substitution list.

Scoped to **HR Dashboard** only. Structured for a hiring manager confirming hands-on experience.

Evidence base. The packaged workbook **HR Analytics Dashboard.twb**, parsed from its XML, plus a short set of answers the author gave directly. The two are kept clearly apart throughout. Where neither supplies an answer, the entry reads **"Not recorded"** and the question appears in section 12.

Confirmed by the author, not by the file:

Item	Answer
Author	Atharva Devne
Why it was built	Self-directed, for the author's own skill development. Not a course, not a client, not an employer.
Dataset origin	Kaggle
Dataset public?	Yes. This resolves the only disclosure concern the file raised, since the workbook embeds full extracts and publishes to Tableau Public.

Scope note. This is an individual Tableau dashboard build over one local flat file. It is not pipeline, warehouse or platform work, and this brief does not pretend otherwise. Sections 2, 4, 6, 8 and 10 are therefore short, and the reasons are stated rather than padded.

1. PROJECT CONTEXT

Item	Detail
Business problem	No stated problem, but unlike Dashboard 1 this workbook does contain one piece of authored prose: a dashboard title text object reading HR ANALYTICS DASHBOARD . What the build addresses, read from its structure: Attrition Count is the measure behind 5 of the 7 worksheets and 4 of the 7 worksheet names contain "Attrition", so it is an attrition concentration view. Where do leavers cluster, by department, education field, gender and age band?
Who requested it	Nobody. Self-directed.
What decisions it supports	No external decision. The purpose was the author's own skill development. What it mechanically enables a reader to ask is in Part A section 16.
My role	Sole author and sole builder. Connection, extract, semantic layer, metric definition, parameter, 7 worksheets, dashboard layout, filter scoping, six actions, tooltip authoring, publication.
Who else was involved	Nobody. No collaborators, no reviewers. Every data source is <code>inline='true'</code> and every connection points at one local file on one machine.
Timeline	Not recorded.
Tool	Tableau Desktop, <code>source-build='2025.2.0 (20252.25.0514.2217)'</code> , <code>source-platform='mac'</code> , document format <code>version='18.1'</code> .
Current status	Published to Tableau Public. <code>repository-location</code> id <code>HRAnalyticsDashboard_17561606616710</code> , dashboard path <code>/workbooks/HRAnalyticsDashboard_17561606616710/HRDashboard</code> , derived from <code>?rev=1.0</code> , current <code>revision='1.1'</code> , so at least two publish revisions. Whether it is live right now could not be checked: the network policy in the documenting environment denied <code>public.tableau.com</code> .

2. SOURCE SYSTEMS AND DATA INVENTORY

There is no source system in the enterprise sense. No database, no API, no SharePoint, no shared folder, no server. Local files on one Mac.

Four data sources are declared. Only one is used.

#	Caption	Internal name	Connector	File	Status
1	HR data (HR Data)	federated.0f0h03plit3gvo176rhln1y1tx4v	textscan (active), plus a declared but unbound excel-direct connection	HR_data.csv	ACTIVE. All 7 worksheets and the dashboard bind to it.
2	HR Data.xlsx - HR data	federated.049m4yglpsanb61b4tu3001365fp	textscan	HR Data.xlsx - HR data.csv	UNUSED, extract still present
3	HR data	federated.lg91kqr1twrfkg14hluc71es07jw	textscan	HR_data.csv	UNUSED, extract still present
4	HR data (2)	federated.1u260d51era62r18fltmd02kg202	textscan	HR_data.xlsx	UNUSED, extract still present

Plus a fifth `<datasource>` named `Parameters` with `hasconnection='false'`, which is Tableau's container for the one parameter.

All four sit in `/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main` and **all four declare the identical 39-column schema**.

Consequence for the packaged file: all four have extracts enabled, which is why the archive carries **four** `.hyper` files of 196,608 bytes each, three of them dead weight.

The unbound connection, transcribed exactly: `class='excel-direct', cleaning='no', compat='no', dataRefreshTime='', filename='/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main/HR_data.csv', interpretationMode='0', validate='no', workgroup-auth-mode='as-is'`. That is the Excel connector pointed at a `.csv` path. It is declared but the `<relation>` binds to the `textscan` connection `textscan.0ry86yc0w1h17j1d0atub17ll83k` instead.

Attribute	Value (active source)
Relation	<code><relation name='HR_data.csv' table='[HR_data#csv]' type='table' /></code>
Schema	39 columns, flat, no key declared
Columns consumed	15 of 39
Provenance	Kaggle, public dataset. Confirmed by the author.
Owner	Publicly available, no internal owner
Extract	<code>enabled='true', count='-1', units='records', with <relation name='Extract' table='[Extract].[Extract]' type='table' /> under <properties context='extract'></code>
Data source filter	One, on <code>[Education]</code> , at <code><shared-view></code> scope
Legacy versus current	Not applicable. Four variants of one file, no history, no second system, no date column.

The 39 source columns

#	Column	Type	#	Column	Type
0	<code>Attrition</code>	string	20	<code>Environment Satisfaction</code>	integer
1	<code>Business Travel</code>	string	21	<code>Hourly Rate</code>	integer
2	<code>CF_age band</code>	string	22	<code>Job Involvement</code>	integer
3	<code>CF_attrition label</code>	string	23	<code>Job Level</code>	integer
4	<code>Department</code>	string	24	<code>Job Satisfaction</code>	integer
5	<code>Education Field</code>	string	25	<code>Monthly Income</code>	integer
6	<code>emp no</code>	string	26	<code>Monthly Rate</code>	integer
7	<code>Employee Number</code>	integer	27	<code>Num Companies Worked</code>	integer
8	<code>Gender</code>	string	28	<code>Percent Salary Hike</code>	integer
9	<code>Job Role</code>	string	29	<code>Performance Rating</code>	integer

#	Column	Type	#	Column	Type
10	Marital Status	string	30	Relationship Satisfaction	integer
11	Over Time	string	31	Standard Hours	integer
12	Over18	string	32	Stock Option Level	integer
13	Training Times Last Year	integer	33	Total Working Years	integer
14	Age	integer	34	Work Life Balance	integer
15	CF_current Employee	integer	35	Years At Company	integer
16	Daily Rate	integer	36	Years In Current Role	integer
17	Distance From Home	integer	37	Years Since Last Promotion	integer
18	Education	string	38	Years With Curr Manager	integer
19	Employee Count	integer			

Three columns arrive pre-derived. `CF_age band`, `CF_attrition label` and `CF_current Employee` carry the `CF_` prefix Tableau uses when a calculated field is exported to a flat file. **They carry no calculation in this workbook**, so that derivation happened upstream. `CF_age band` is used directly on a shelf. How they were produced is **not recorded**.

24 of the 39 columns are never used on any shelf, filter or calculation: `Business Travel`, `Marital Status`, `Over Time`, `Over18`, `Training Times Last Year`, `Daily Rate`, `Distance From Home`, `Environment Satisfaction`, `Hourly Rate`, `Job Involvement`, `Job Level`, `Monthly Income`, `Monthly Rate`, `Num Companies Worked`, `Percent Salary Hike`, `Performance Rating`, `Relationship Satisfaction`, `Standard Hours`, `Stock Option Level`, `Total Working Years`, `Work Life Balance`, `Years At Company`, `Years In Current Role`, `Years Since Last Promotion`, `Years With Curr Manager`.

3. SOURCE-TO-TARGET MAPPING

Target field	Source field	Transformation rule	Notes
Attrition Count	Attrition (string Yes/No)	IF [Attrition] = 'Yes' THEN 1 ELSE 0 END	The only conditional logic in the workbook. Converts text to a summable flag.
Attrition Rate	Attrition Count, Employee Count	SUM([Calculation_231935388807032832])/SUM([Employee Count])	Aggregate over aggregate, so it recomputes at the view's level of detail
Active Employees	Employee Count, Attrition Count	SUM([Employee Count]) - SUM([Calculation_231935388807032832])	Spacing transcribed exactly as stored
Age (bin)	Age, [Age Parameter]	class='bin', formula='[Age]', peg='0', decimals='0', size-parameter='[Parameters].[Age Parameter]'	Bin width is parameter-driven, not fixed
Number of Records	none	1, user:auto-column='numrec'	Auto-generated by Tableau, never used

Target field	Source field	Transformation rule	Notes
min(1)	none	min(1) , ad hoc inside one worksheet	Dual-axis placeholder
Department1	Department	Caption only	Collision artefact, not a business rename
Education1	Education	Caption only	
Education Field1	Education Field	Caption only	
Gender1	Gender	Caption only	
Job Role1	Job Role	Caption only	
Job Satisfaction1	Job Satisfaction	Caption; role measure → dimension ordinal , aggregation='Sum' retained	
Emp No	emp no	Caption only	Declared, never used on a shelf
Employee Number	Employee Number	Role measure → dimension ordinal	Declared, never used on a shelf
CF age band	CF_age band	Caption only, underscore removed. No derivation here.	Used directly on Columns
CF attrition label	CF_attrition label	Caption only	Declared, never used
CF current Employee	CF_current Employee	Caption only	Declared, never used
Employee Count	Employee Count	No override at all	Used as SUM([Employee Count]) , the denominator of the rate

Fields whose meaning differed between systems. Not applicable, one system. The closest real analogue is **within** the workbook: `Department` carries a `Department1` caption and two auto action groups (`Action (Department)` and `Action (Department1)`) that both resolve to the same underlying `[Department]` level. That is a Tableau artefact of a data source replacement, not a cross-system semantic mismatch.

Grain

Object	Grain
Source	One row per employee. Inferred from the schema: two identifier columns plus an <code>Employee Count</code> column.
KPI	Whole filtered population
Attrition by Gender	One mark per gender
Department wise Attrition	One mark per department
No. of Employee by Age Group	One mark per age bin
Job Satisfaction Rating	One mark per job role x satisfaction score
Education Field wise Attrition	One mark per education field
Attrition Rate by Gender for Different Age Group	One mark per age band x gender

Grain mismatches: none. Every target aggregates upward from a single-grain source. No join exists anywhere, so no fan-out and no double-counting risk.

4. DATA FLOW AND ARCHITECTURE

No orchestration tool, no staging layer, no warehouse, no lakehouse, no scheduler.



Stage	Where data lands
Source	Local file system
Staging	None
Warehouse / lakehouse	None
Curated files	Four Hyper extracts inside the packaged workbook, only one referenced
Serving	Tableau Public

Inherited or previous architecture: not recorded. Two artefacts hint at prior work without evidencing an architecture: the three unused data sources, which show the same dataset imported more than once in different formats, and the three `CF_` columns, which show that some derivation happened before this workbook.

Flow in boxes and arrows: upstream derivation (unrecorded) → local delimited files → Tableau Desktop textscan connector → four Hyper extracts → semantic layer (captions, roles, parameter, parameter-driven bin, calculations, data source filter) → 7 worksheets → 1 dashboard over a background image → Tableau Public. Two side boxes feed the dashboard: "six filter actions" and "one parameter slider", and the parameter side box also feeds back into the semantic layer box, because it drives the bin definition.

5. DATA CONSOLIDATION, CLEANING AND STANDARDISATION

No consolidation occurred. Four data sources exist, which is the usual precondition for blending, but no worksheet references more than one. No join, no union, no blend, no linking field. The other three sources are simply inert.

Every cleaning rule applied, all ten

#	Rule	Exact definition	Fields affected	Layer	Why there
1	Text flag to numeric indicator	<code>IF [Attrition] = 'Yes' THEN 1 ELSE 0 END</code>	new <code>Attrition Count</code> , from <code>Attrition</code>	Calculated field	The source column is the string <code>Yes/No</code> and cannot be aggregated
2	Rate derivation	<code>SUM([Calculation_231935388807032832])/SUM([Employee Count])</code>	new <code>Attrition Rate</code>	Calculated field	Written as aggregate over aggregate so it recomputes at any level of detail
3	Complement derivation	<code>SUM([Employee Count]) - SUM([Calculation_231935388807032832])</code>	new <code>Active Employees</code>	Calculated field	
4	Parameter-driven binning	<code>class='bin', formula='[Age]', peg='0', decimals='0', size-parameter=[Parameters].[Age Parameter]</code>	new <code>Age (bin)</code>	Calculated field bound to a parameter	Lets the viewer change granularity at view time rather than committing to one bucket width
5	Role reclassification	<code>role='dimension' type='ordinal'</code>	<code>Job Satisfaction</code> , <code>Employee Number</code>	Data source	A 1-to-4 satisfaction score is a category, not a quantity to sum
6	Display renaming	<code>caption</code> attribute	10 fields	Data source	6 of these are <code>Name1</code> collision artefacts rather than deliberate renames
7	Global scope filtering	<code><shared-view> filter on [none:Education:nk], function='level-members', ui-enumeration='all'</code>	<code>Education</code>	Data source scope	Applies to every worksheet using the data source, including the KPI sheet
8	Presentation-layer label correction	Hand-typed labels inside <code><customized-tooltip></code>	<code>Department:</code> , <code>Gender:</code> , <code>Job Role:</code> , <code>Education Field:</code> , <code>Job Satisfaction:</code>	Worksheet	Corrects the <code>Name1</code> captions where the user actually sees them

#	Rule	Exact definition	Fields affected	Layer	Why there
9	Explicit category ordering	<manual-sort column='[none:CF_age band:nk]' direction='ASC'> with dictionary "Under 25", "25 - 34", "35 - 44", "45 - 54", "Over 55"	CF_age band	Worksheet	Alphabetical ordering would place Over 55 in the middle
10	Explicit measure ordering	<manual-sort column='[:Measure Names]' direction='ASC'> with a 4-member dictionary	Measure Names on KPI	Worksheet	Pins KPI tile order

Cleaning rules NOT applied

Rule category	Status
Null handling (ISNULL , IFNULL , ZN)	Not present
Duplicate detection or removal	Not present
Explicit type casting (INT() , STR() , FLOAT() , DATE())	Not present. The IF/THEN changes the encoding but is not a cast function.
Date parsing, date parts, timezone handling	Not present, and not possible. No date column exists. Years At Company , Years In Current Role , Years Since Last Promotion , Years With Curr Manager and Total Working Years are integer tenure counts, not dates.
Text normalisation (TRIM , UPPER , REPLACE , REGEXP)	Not present
Value aliases	Not present. No <aliases> element anywhere.
Outlier detection or capping	Not present
Country, currency or unit standardisation	Not present
Domain classification logic	Not present in this workbook. It happened upstream in the CF_ columns.
Referential validation	Not present. No joins to validate.
Groups (user-created)	Not present. The four <group> elements are auto-generated hidden action groups (user:auto-column='sheet_link'), not data groups.
Sets, hierarchies, drill paths	Not present
Extract filter	Not present
Tableau Prep flow	Not present in the archive

Which layer performed each transformation, and why there

All of it was done inside Tableau Desktop. No SQL, no Python, no Prep flow is referenced. The three **CF_** columns are the one piece of evidence that derivation happened before Tableau, but how is **not recorded**.

Layer	What sits there	Why
Data source scope	Captions, role conversions, the Education shared-view filter	Inherited by every worksheet, including the KPI sheet
Calculated field scope	Attrition Count , Attrition Rate , Active Employees , Age (bin)	Reusable business metrics, and Tableau enforces the dependency chain

Layer	What sits there	Why
Worksheet scope	<code>min(1)</code> placeholder, two manual sorts, all tooltips and labels	Local to one view
Dashboard scope	Education dropdown, <code>Age Size</code> slider, six filter actions	Interaction, not data

6. DATA MODEL: SCHEMA AND TABLE DESIGN

No dimensional model was designed. This is the single most important scope boundary to understand about this artefact.

Design element	Status
Fact tables	None
Dimension tables	None
Star or snowflake schema	None
Surrogate key strategy	None. <code>Employee Number</code> and <code>emp no</code> exist as natural identifiers but are never used on any shelf, filter or calculation.
Slowly changing dimensions	None
Date dimension	None, and impossible. No date column exists.
Relationships and filter directions	None. Four single-table sources, never combined.
Historical migration or backfill	None
Cutover between systems	Not applicable
Cardinality and referential integrity settings	Not applicable

Design rules that WERE set, and the reason for each

#	Rule	Evidence	Reason
1	Extract everything, filter nothing at extract time	<code>count='-1'</code> on all four sources, no extract filter	Keeps the workbook self-contained and keeps all filtering interactive
2	Embed the data source rather than publish it	<code>inline='true'</code>	Self-contained portfolio artefact
3	Ordinal scores are dimensions, not measures	<code>role='dimension' type='ordinal'</code> on <code>Job Satisfaction</code> and <code>Employee Number</code>	A 1-to-4 score is a category; an identifier should never be summed
4	Define metrics as promoted calculated fields, not per-view expressions	<code>Attrition Count</code> , <code>Attrition Rate</code> , <code>Active Employees</code> at data source level	One definition per business metric, reused across sheets
5	Write ratios as aggregate over aggregate	<code>SUM(...)/SUM(...)</code>	So the ratio recalculates correctly at any level of detail rather than averaging row-level ratios

#	Rule	Evidence	Reason
6	Make analytical granularity a user choice	<code>size-parameter='[Parameters].[Age Parameter]'</code>	Avoids committing to one bucket width, and lets a viewer test whether a pattern survives re-bucketing
7	Scope global filters at the data source	<code><shared-view></code> filter on <code>[Education]</code>	Guarantees the KPI sheet responds, which a hand-picked per-sheet filter selection can miss
8	Bind dashboard titles to sheet names	<code><Sheet Name></code> token in 6 titles	Renaming a worksheet renames its dashboard title automatically
9	Sequential colour for quantities, categorical for categories	<code>blue_10_0</code> and <code>purple_10_0</code> interpolated ramps on <code>SUM(Employee Count)</code> ; default categorical on <code>Gender</code> and <code>Department</code>	The encoding matches the data type

7. BUSINESS RULES, VALIDATION AND GOVERNANCE

Business rules encoded

Three, all real and all consequential.

#	Rule	Exact definition	Consequence
1	What counts as attrition	<code>IF [Attrition] = 'Yes' THEN 1 ELSE 0 END</code>	Attrition is whatever the source column marks <code>Yes</code> . There is no voluntary versus involuntary split, no tenure qualification, no status exclusion and no date window.
2	What counts as active	<code>SUM([Employee Count]) - SUM([Attrition Count])</code>	The population is a closed set: every employee is either active or a leaver. A snapshot, not a point-in-time headcount, because no hire or termination date exists.
3	How attrition rate is defined	<code>SUM([Attrition Count])/SUM([Employee Count])</code>	Leavers over total headcount in the current view scope. No annualisation, no average-headcount denominator.

Whether the source system's own definition of `Attrition = 'Yes'` includes involuntary exits: not recorded.

Validation checks

None are present. A complete absence, not an omission from this brief.

Check type	Present?
Reconciliation to source	No
Row or control totals	No
Cross-system checks	Not applicable, one system
Step-change or regression checks	No
Data-quality flags shown to users	No

Whether validation was performed outside Tableau is **not recorded**.

Governance

Item	Status
Metric definitions and data dictionary	None as a document. The calculated field names (<code>Attrition Count</code> , <code>Attrition Rate</code> , <code>Active Employees</code>) are self-describing, which is the closest thing present.
Sign-off process and owners	None . Self-directed work.
Decision log, open questions register, documentation	None
Version control	Tableau Public revision counters only: <code>revision='1.1'</code> , derived from <code>?rev=1.0</code> , so at least two publishes
Circumstantial repository marker	The source folder is named <code>HR-Analytics-Dashboard-Using-Tableau-main</code> . The <code>-main</code> suffix is what downloading a GitHub repository ZIP of a <code>main</code> branch produces, so a repository was involved somewhere in the project's history. Whose and for what: not recorded .

Defects observable in the delivered artefact

Found by reading the XML, not reported by anyone. All nine are **unfixed in the delivered file**.

#	Defect	Evidence
1	Three of four data sources are unused but still carry extracts, adding three redundant <code>.hyper</code> files to the packaged archive	Archive listing plus worksheet dependencies
2	Six field captions are <code>Name1</code> collision artefacts (<code>Department1</code> , <code>Gender1</code> , <code>Job Role1</code> , <code>Education1</code> , <code>Education Field1</code> , <code>Job Satisfaction1</code>)	<code>caption</code> attributes. Partly mitigated : tooltips hand-type the clean labels, but legends still show the artefacts.
3	The worksheet titled <code>Attrition Rate by Gender for Different Age Group</code> does not use the <code>Attrition Rate</code> field . Its Angle shelf carries <code>SUM([Attrition Count])</code> and its labels carry a percent-of-total table calculation.	Shelf contents and tooltip text
4	No number formatting is defined, so <code>Attrition Rate</code> renders as a decimal rather than a percentage	No <code>number-format</code> attribute in the file
5	The workbook was saved with a <code>Department = "R&D"</code> mark selected, persisting an action filter into 6 of 7 worksheets	<code><groupfilter function='member' member=' "R&D" ' user:ui-action-filter='[Action1]' /></code>
6	<code>Job Satisfaction Rating</code> has a continuous colour encoding but no colour legend on the dashboard	Dashboard zone list
7	<code>Education Field wise Attrition</code> has no sort , so bars appear in data source order rather than ranked	No <code><sort></code> or <code><manual-sort></code> on that sheet
8	The dashboard name carries a trailing space : <code>HR Dashboard</code>	<code><dashboard name='HR Dashboard '></code>
9	The parameter has three different names in three places: internal <code>Age Parameter</code> , caption <code>Bin Size</code> , control title <code>Age Size</code>	Parameter definition and zone title

Defect 5 is the one to check before any demo. The saved state carries a live R&D filter on six of seven views, so any number read off the dashboard in that state is an R&D number, not an organisation-wide number.

8. REFRESH, OPERATIONS AND SECURITY

Item	Status
Refresh frequency	Not recorded. No schedule, no incremental configuration, no refresh definition exists.
Full or incremental	Full, by construction. All four extracts are <code>count='-1' units='records'</code> with no incremental refresh key and no extract filter.
Refresh trigger	Local file connections, so a refresh requires the file to be present on that Mac and a manual refresh to be run. <code>dataRefreshTime=''</code> on the one connection that declares the attribute.
How the source connects to Tableau	Direct local file read via the <code>textscan</code> connector, then extract. No server, no gateway, no DSN, no connection string, no credentials, no DirectQuery equivalent. The live-versus-extract choice is extract , on all four sources.
Credentials handling	<code>workgroup-auth-mode='as-is'</code> on every connection, the default for a file connection needing no authentication. No credential is stored.
Access control	Published to Tableau Public , public by default with no user-based access control.
Row-level security	None. No user filter, no <code>USERNAME()</code> , no <code>ISMEMBEROF()</code> , no user function appears anywhere.
Workspace administration	Not applicable. Tableau Public has no workspace model.
What data was allowed to leave the source	No restriction, and none needed. The workbook is packaged with full extracts embedded, so publishing it publishes the underlying records, downloadable by any viewer. The dataset is a public Kaggle dataset , confirmed by the author, so there is no disclosure issue. Worth stating explicitly: embedding a full extract in a public workbook would be a real problem with any non-public source, and knowing that distinction is the point.
Monitoring, failure handling, notification	None. No alert, subscription or notification is configured.

9. REQUIREMENTS TO KPIS AND DASHBOARD LOGIC

How requirements were gathered

No requirements were gathered. Self-directed work with no stakeholder, no brief and no session.

KPI list

All five live in **one** worksheet, `KPI`, built with Measure Names and Measure Values.

KPI	Business definition	Decision it supports	Grain	Source fields	Calculation logic	Owner	Sign-off
<code>Employee Count</code>	Headcount in the filtered population	Denominator context for everything else	Current filter scope	<code>Employee Count</code>	<code>SUM([Employee Count])</code>	Auth or	None, self-directed

KPI	Business definition	Decision it supports	Grain	Source fields	Calculation logic	Owner	Sign-off
Attrition Count	Number of leavers in the filtered population	Absolute attrition volume	Current filter scope	Attrition	SUM(IF [Attrition] = 'Yes' THEN 1 ELSE 0 END)	Auth or	None
Attrition Rate	Leavers divided by headcount	Whether attrition is high relative to population size, comparably across slices of different sizes	Current filter scope, recomputed at view level of detail	Attrition, Employee Count	SUM([Attrition Count])/SUM([Employee Count])	Auth or	None
Active Employees	Headcount remaining after attrition	Retained population	Current filter scope	Employee Count, Attrition	SUM([Employee Count]) - SUM([Attrition Count])	Auth or	None
AVG(Age)	Mean employee age	Flags whether a cross-filter has isolated an unusually young or old cohort	Current filter scope	Age	AVG([Age])	Auth or	None

All five respond to the global Education filter and to all six cross-filter actions, because the filter is data source scoped.

The **KPI** sheet's Measure Names filter is a `function='union'` with `op='manual'` over five explicit members, in this stored order: `SUM(Employee Count)`, `SUM(Attrition Count)`, `Attrition Rate`, `Active Employees`, `AVG(Age)`. A separate manual-sort dictionary lists only the first four, so `AVG(Age)` sorts after them.

Measures written, actual code

No DAX and no SQL exists. The Tableau code, transcribed exactly:

```

// Attrition Count
IF [Attrition] = 'Yes' THEN 1 ELSE 0 END

// Attrition Rate
SUM([Calculation_231935388807032832])/SUM([Employee Count])
// reads in the UI as: SUM([Attrition Count])/SUM([Employee Count])

// Active Employees
SUM([Employee Count]) - SUM([Calculation_231935388807032832])
// reads in the UI as: SUM([Employee Count]) - SUM([Attrition Count])

// Number of Records (auto-generated, unused)
1

// min(1) (ad-hoc dual-axis placeholder)
min(1)

// Age (bin) (a bin definition, not a formula)
class = bin, formula = [Age], peg = 0, decimals = 0,
size-parameter = [Parameters].[Age Parameter]

// Parameter
[Age Parameter], caption "Bin Size", real, param-domain-type = range,
value = 3.0, min = 2.0, max = 10.0, granularity = 1.0

// The one table calculation, applied on 2 sheets
<table-calc ordering-type='Rows' type='PctTotal' />
// surfaced in tooltips as: "% of Total Attrition Count along Table (Across)"

```

There are no LOD expressions. Verified by absence of `{ }` in every formula.

Dependency chain



Dashboard design

Item	Value
Pages	1 dashboard, no story, no navigation
What it shows	Header with a title text object and the Education filter. KPI band with 5 tiles plus a gender lollipop. Middle band with a department pie, a parameterised age histogram and a job-role-by-satisfaction highlight table. Bottom band with an education field bar chart and a row of age band donuts split by gender.
Visuals used	Measure Names/Values KPI strip, lollipop (Bar + Circle dual axis), pie with percent of total, parameterised bar histogram with a sequential purple ramp, highlight table with Square marks and a sequential blue ramp, horizontal bar, donut small multiples with the band total printed in the hole
Filters shown to the user	1 dropdown, Education , applied to all 7 sheets via data source scope
Parameter controls	1 slider, Age Size , real, 2.0 to 10.0, step 1.0, default 3.0
Drill paths	None. No hierarchy exists. Cross-filtering by clicking substitutes for drill-down.
How assumptions and definitions are shown to the user	Partially. Custom tooltips on all 7 sheets spell out field labels and name the table calculation explicitly as % of Total Attrition Count along Table (Across) . There is no definition of what Attrition = 'Yes' means and no assumptions panel.
Canvas	1580 x 900, sizing-mode='fixed' , floating objects over Image/HR background.pptx.png

Interactivity

Six filter actions, one sourced from every worksheet except **KPI**, all **command='tsc:tsl-filter'**, all **on-select** with **auto-clear='true'**, all **special-fields='all'**, all targeting the whole dashboard. Every chart is a cross-filter surface and the KPI strip recomputes on every click.

Anything the business explicitly asked for in the presentation: nothing. There was no business.

10. STAKEHOLDER COORDINATION

Nothing to report, and that is accurate rather than a gap in the evidence. The project was solo and self-directed. There were no stakeholders, no meetings, no working sessions, no agreements, no pushback and no disagreements to handle. The workbook contains no name, comment, review or approval artefact of any kind.

11. OUTCOMES AND STATUS

Item	Status
What is live	Published to Tableau Public, workbook id <code>HRAnalyticsDashboard_17561606616710</code> , dashboard repository id <code>HRDashboard</code> . Published at least twice (revision advanced <code>1.0</code> → <code>1.1</code>). Whether currently live could not be checked, the documenting environment's network policy denied <code>public.tableau.com</code> .
Completeness	Complete build. All 7 worksheets in the workbook are placed on the dashboard. No orphaned, unfinished or staged worksheets.
What is validated	Nothing. No validation artefact exists.
What is signed off	Not applicable, self-directed.
What is still open	The 9 defects in section 7, all unfixed. The three most consequential: the rate-versus-count title mismatch, the persisted <code>R&D</code> selection, and the absence of percentage formatting on the one rate KPI.

Measurable results

None can be claimed. No view count, adoption figure, time saving, grade, feedback or business outcome exists. No data values were read either, because the `.hyper` extracts are binary and no Hyper reader was available, so no headcount, attrition rate or departmental ranking can be stated.

What can be counted, and only this:

Metric	Value
Worksheets built	7
Worksheets placed on the dashboard	7 of 7
Calculated fields authored	6
Parameters authored	1

Metric	Value
Calculation dependency depth	3 levels
Source columns consumed	15 of 39
Distinct chart forms	7
Dual-axis constructions	2
Table calculations	1 type, on 2 sheets
Manual sort dictionaries	2
Dashboard actions	6
Custom tooltip definitions	8, across 7 worksheets
Custom mark label definitions	6
Colour legends on the dashboard	3
Publish revisions	at least 2

Lessons learned

Not recorded in the workbook. The defect list in section 7 is an external reading of the file.

12. GAPS AND QUESTIONS

Already answered

Question	Answer
Was this for a course, a client, or self-directed?	Self-directed, for your own development.
Where did the dataset come from?	Kaggle.
Is the dataset public, given the embedded extracts?	Yes, public. No disclosure issue.

Still open

1. **How were the `CF_age band`, `CF_attrition label` and `CF_current Employee` columns derived, and by whom?** Were they already in the file you downloaded, or did you create them in an earlier Tableau workbook and export?
2. **What does `Employee Count` actually contain?** If it is 1 on every row, confirm it, because it is the denominator of your only rate KPI.
3. **Was the CSV cleaned before it reached Tableau?** If yes, with what tool and what exact rules?

4. **Does the source define `Attrition = 'Yes'` to include involuntary exits?**
 5. **Why does the workbook have four data sources?** What did you try first, and why did `HR_data.csv` win?
 6. **Was the bottom chart originally intended to show `Attrition Rate` and then changed to `Attrition Count`, or was the title always approximate?**
 7. **Were the `Department1`, `Gender1`, `Job Role1` captions intentional**, or collision artefacts from replacing a data source?
 8. **Was the saved `R&D` selection deliberate?**
 9. **Why six filter actions rather than a highlight action?** A strong answer on the difference lands well.
 10. **Where did `HR background.pptx.png` come from?** The filename says PowerPoint. Did you design it?
 11. **Did anyone review or use it**, and what did they say?
 12. **The confirmed live Tableau Public URL.** The recoverable id is `HRAalyticsDashboard_17561606616710`.
 13. **Any view counts or feedback?**
 14. **What changed between the two publish revisions?**
 15. **One concrete insight you personally found** in this dataset while building. Not a figure the dashboard computes, but something you noticed.
 16. **Which of the 9 defects do you agree with**, and does any have an explanation this reading has missed?
-

13. SANITISED VERSION

The same brief with identifying details replaced. All technical detail, counts, rules and logic unchanged.

1. Project context. Self-directed work by a single author for their own skill development, not for a course, client or employer. No requester, no stakeholder, no collaborators. Subject matter: workforce attrition analysis, evidenced by an attrition measure behind 5 of 7 worksheets and 4 of 7 worksheet names containing "Attrition". The workbook contains one piece of authored prose, a dashboard title text object. BI tool version 2025.2.0 on a workstation. Published to a public visualisation hosting platform, at least twice.

2. Source systems and data inventory. No enterprise source system. Four data sources declared, all pointing at the same folder on one workstation and all declaring the identical 39-column schema, plus a fifth parameter container with no connection. Only one is used; the other three are inert but still carry their own extracts, which is why the packaged archive holds four in-memory extract files of which three are dead weight. The active source declares two connections, a delimited text connection that the relation binds to and a spreadsheet connection that it does not, the latter with `cleaning='no'`, `compat='no'`, `interpretationMode='0'`, `validate='no'`. Provenance: a public dataset from a public data-sharing platform. Schema covers an attrition flag, headcount, organisational unit, education field and level, gender, job role, five satisfaction and involvement scores, seven compensation and rate columns, five tenure columns, employee identifiers and

demographics. Three columns arrive pre-derived carrying the tool's exported-calculated-field prefix, with no calculation in this workbook, so that derivation happened upstream by means not recorded. 15 of 39 columns are consumed; the other 24 are named in full in the unsanitised section 2.

3. Source-to-target mapping. Nineteen target fields mapped, covering one conditional indicator, one aggregate ratio, one aggregate difference, one parameter-bound bin, one auto-generated constant, one dual-axis placeholder, six captioned dimensions (all six captions being collision artefacts), two role-converted identifiers and scores, three pre-derived columns consumed as-is, and one headcount column used with no override at all. No cross-system semantic mismatch existed; the closest analogue is an internal one where a single underlying field carries a duplicate caption and two auto action groups after a data source replacement. Source grain is one row per employee; target grains range from whole-population to one mark per age band and gender. No grain mismatches, because no join exists.

4. Data flow. Upstream derivation (unrecorded) → local delimited files → BI desktop tool, delimited text connector → four columnar in-memory extracts, all records, no filter → semantic layer (captions, roles, parameter, parameter-driven bin, calculations, data-source-scoped filter) → 7 worksheets → one dashboard over a background image → public hosting platform. No staging, no warehouse, no lakehouse, no scheduler, no SQL step, no scripting step, no ETL tool. Two side boxes feed the dashboard, six filter actions and one parameter slider, and the parameter also feeds back into the semantic layer because it drives the bin.

5. Cleaning and standardisation. No consolidation: four sources exist but no worksheet references more than one, so no join, union, blend or linking field. Ten rules, transcribed in full in the unsanitised section 5: text flag to numeric indicator; rate derivation as aggregate over aggregate; complement derivation; parameter-driven binning; role reclassification on two columns; display renaming on ten fields; global scope filtering via a shared-view filter; presentation-layer label correction inside every custom tooltip; explicit category ordering via a manual sort dictionary running from the youngest band to the oldest; and explicit measure ordering. Not applied: null handling, duplicate detection, explicit type casting, date parsing or timezone handling (impossible, no date column), text normalisation, value aliases, outlier treatment, unit or code standardisation, domain classification in this workbook, referential validation, user-created groups, sets, hierarchies, extract filter, external preparation flow. All work was done inside the BI tool across four deliberate layers.

6. Data model. No dimensional model. No fact or dimension tables, no surrogate keys (two natural identifiers exist but are never used), no slowly changing dimensions, no date dimension (impossible), no relationships, no migration, no cutover. Nine design rules were set and are listed with reasons, including defining metrics as promoted calculated fields, writing ratios as aggregate over aggregate, making analytical granularity a user choice, scoping global filters at the data source, binding dashboard titles to sheet names, and matching colour encoding to data type.

7. Business rules, validation, governance. Three business rules, all consequential: what counts as attrition (a single text-equality test, with no voluntary versus involuntary split, no tenure qualification, no status exclusion and no date window); what counts as active (headcount minus leavers, a snapshot rather than a point-in-time figure because no hire or termination date exists); and how the rate is defined (leavers over headcount in the current view scope, with no annualisation and no

average-headcount denominator). No validation checks of any kind. No data dictionary, sign-off, decision log or documentation; version control limited to publish revision counters, with a folder-name suffix indicating a repository was involved somewhere. Nine defects are verifiable and unfixed: three unused data sources carrying extracts; six collision-artefact captions; a worksheet titled as a rate that plots a count; no number formatting, so the rate renders as a decimal; a persisted mark selection saved into six of seven worksheets; a continuous colour encoding with no legend exposed; an unsorted bar chart; a dashboard name with a trailing space; and a parameter carrying three different names in three places.

8. Refresh, operations, security. Refresh frequency not recorded. Full by construction on all four extracts. Direct local file read then extract, with no server, gateway, DSN, connection string, credentials or direct-query equivalent. Authentication mode is the default for a file connection needing none. Published to a public-by-default platform with no user-based access control and no row-level security; no user function appears anywhere. The workbook embeds full extracts, so publishing it publishes the records, but the dataset is public so there is no disclosure issue. No monitoring, failure handling or notification.

9. Requirements to KPIs. No requirements were gathered; the work was self-directed. Five KPIs in a single worksheet built with the tool's measure-name and measure-value constructs: headcount as a sum; leaver count as a sum of the conditional indicator; a rate as the aggregate ratio; active headcount as the aggregate difference; and a mean age. All five respond to the global filter and to all six cross-filter actions. No DAX and no SQL; the actual code is one conditional indicator, one aggregate ratio, one aggregate difference, one auto-generated constant, one dual-axis placeholder, one parameter-bound bin definition, one range parameter and one built-in percent-of-total table calculation, all transcribed unchanged in the full section 9. The dependency chain runs three levels deep from a source text column. No level-of-detail expressions. Dashboard: one page on a 1580 x 900 fixed canvas with floating objects over a background image, a header with a title object and the global filter, a KPI band of five tiles plus a gender lollipop, a middle band with an organisational-unit pie, a parameterised age histogram and a role-by-satisfaction highlight table, and a bottom band with an education field bar chart and a row of age band donuts. One dropdown on all seven sheets, one slider parameter ranging 2.0 to 10.0 by 1.0 defaulting to 3.0, no drill paths, and six on-select auto-clear filter actions passing all fields. Assumptions are partially shown: every worksheet carries a hand-written tooltip and the table calculation is named explicitly, but the attrition rule itself is never defined to the user.

10. Stakeholder coordination. None. Solo, self-directed work with no stakeholders, meetings, agreements or disagreements.

11. Outcomes and status. Published at least twice. Complete build: all 7 worksheets placed on the dashboard, none orphaned. Nothing validated, nothing signed off, nine defects still open. No measurable results can be claimed and no data values were read. Countable facts: 7 worksheets, 6 calculated fields, 1 parameter, a 3-level dependency chain, 15 of 39 source columns consumed, 7 distinct chart forms, 2 dual-axis constructions, 1 table calculation type on 2 sheets, 2 manual sort dictionaries, 6 dashboard actions, 8 custom tooltip definitions, 6 custom label definitions, 3 colour legends, at least 2 publish revisions.

12. Gaps and questions. Three answered (self-directed, public data-sharing platform origin, dataset is public). Sixteen still open, listed in full in the unsanitised section 12.

Substitution list

#	Original	Replaced with
1	Tableau, Tableau Desktop, Tableau Public	"BI desktop tool", "public visualisation hosting platform"
2	Hyper, .hyper	"columnar in-memory extract"
3	textscan connector	"delimited text connector"
4	excel-direct connector	"spreadsheet connector"
5	Measure Names, Measure Values	"the tool's measure-name and measure-value constructs"
6	Atharva Devne	"a single author" / "the author"
7	atharvadevne (account name in file paths)	omitted
8	/Users/atharvadevne/Desktop/HR-Analytics-Dashboard-Using-Tableau-main	"the same folder on one workstation"; the -main suffix described as "a folder-name suffix indicating a repository was involved"
9	Kaggle	"a public data-sharing platform"
10	HR Analytics Dashboard.twb	"the workbook"
11	HR_data.csv, HR data.csv, HR data.xlsx, HR Data.xlsx - HR data.csv	"four files of the same dataset in different formats and naming conventions"
12	HRAnalyticsDashboard_17561606616710, HRDashboard	omitted
13	public.tableau.com	"the host"
14	HR Dashboard (dashboard name)	"the dashboard"; the trailing-space defect described without reproducing the name
15	Image/HR background.pptx.png	"a background image"; PowerPoint described as "a presentation tool"
16	Department, Department1	"organisational unit"
17	Education, Education1	"education level"
18	Education Field, Education Field1	"education field"
19	Job Satisfaction, Job Satisfaction1	"satisfaction score"
20	Attrition	"attrition flag"

#	Original	Replaced with
2 1	Employee Count	"headcount column"
2 2	Employee Number, emp no	"employee identifiers"
2 3	CF_age band, CF_attrition label, CF_current Employee	"three pre-derived columns"
2 4	R&D (the persisted selection member)	"a persisted mark selection"
2 5	"Under 25" through "Over 55" (sort dictionary)	"running from the youngest band to the oldest"
2 6	Federated data source ids (federated.0f0h03p1it3gvo176rhlnly1tx4v and so on)	omitted
2 7	Internal calculation ids ([Calculation_231935388807032832] and so on)	replaced with the readable field name
2 8	Dashboard UUID {30867C40-EA91-4DA5-AF47-E92ACECA09BD}	omitted
2 9	Extract filenames (#TableauTemp_0uykjkf... and so on)	omitted
3 0	Worksheet names	described by chart type and content
3 1	Palette names blue_10_0, purple_10_0	"sequential ramps"
3 2	macOS	"workstation"

Not substituted, deliberately: all formulas, the parameter range and granularity, canvas dimensions, zone counts, filter scopes, action counts, worksheet counts, column counts and dependency depth, because the instruction was to keep technical detail, counts, rules and logic unchanged.

One item to check yourself: the sanitised version still states the subject is workforce attrition. If the subject matter itself is sensitive, replace it with something like "a workforce dataset".